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Neurogenic Bowel: Dysfunction and Rehabilitation

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SUMMARY PEARLS

- GI dysfunction is prevalent with neurologic diseases affecting the brain, spinal cord, and peripheral nervous system.
- Normal functioning of the GI tract requires coordination of muscle contraction, digestion and absorption of nutrients, preparation and movement of waste, voluntary defecation, and adequate elimination of stool; all regulated by interactions with the brain and spinal cord, autonomic nervous system and enteric nervous system (ENS), and somatomotor system.
- Management of NBD involves a stepwise method that commences with conservative measures including minimizing constipating medications, good intake of fluids and fiber, to utilization of oral laxatives, rectal medications, strategies for bowel evacuation, and can progress to more invasive measures and surgical procedures.
- Interdisciplinary rehabilitative interventions for NBD from the onset of neurologic compromise and throughout life are key components of holistic care. Recognition of all aspects of impairment and disability that limit a person's ability to volitionally defecate and maintain continence must be assessed and addressed to optimize GI function. Prevention and early intervention of more serious medical problems associated with increased clinic and hospital visits are imperative to maintain health, good quality of life, and general well-being.
- Management of NBD symptoms and problems (constipation, fecal incontinence, abdominal and pelvic pain) must be established on strong partnerships between patients, their caregivers, and medical providers to provide personalized care based on real-world factors, self-management tools, and adherence to joint goals of care. This must be reviewed and updated periodically to account for neurodegenerative and age-related effects.

Introduction

Gastrointestinal (GI) dysfunction is most often experienced as a conglomeration of symptoms that indicate lower GI impairment, including constipation, diarrhea, and fecal incontinence (FI). It can also present as upper GI impairment heralded by bloating, nausea, early satiety, burning, and gaseousness. Despite its severity, neurogenic bowel dysfunction (NBD) can be a clinically elusive condition. It is often eclipsed by other, more noticeable associated motor deficits, but can be life-limiting if not thoroughly assessed and treated using rehabilitation principles. Interdisciplinary rehabilitative interventions for NBD focus on establishing a total management plan for bowel function, termed a *bowel program*, and for assisted defecation, known as *bowel care*.^{1,29} Because sensation

and mobility are often restricted, limiting the person's ability to anticipate the need for and to physically perform independent bowel care and hygiene, the overarching rehabilitation approach to this complex, multisystem intersection of impairments must be holistic and have as a goal the design of a bowel program that is liberating and individualized for the patient.

It is surprising that of the many abilities regained during the rehabilitation process, bowel care capabilities at the time of discharge are not always comparable with other skills that would be expected for a given level of function. Bowel continence is one of the greatest predictors of return to home for stroke survivors.³¹ More than one-third of surveyed persons with spinal cord injury (SCI) rated bowel and bladder dysfunction as having the most significant impact on their lives after SCI, and yet bowel management has been found to be one of the areas of least competence among rehabilitated spinal cord-injured persons.⁵² In a Swedish review of medical problems after SCI, 41% of subjects rated bowel dysfunction as a moderately to severely life-limiting

[†]Deceased

problem.⁸⁴ It is obvious that bowel management is an important area that is often undermanaged.

Epidemiology

GI dysfunction is prevalent in patients with neurologic diseases, particularly those who require rehabilitation. In addition to the direct effects of neurologic disease on gut function; debility, insufficient fluid intake, and use of anticholinergics and other medications can play complicating roles in exacerbation of enteric problems. The common digestive tract problems and NBD found in individuals with stroke, brain injury, multiple sclerosis, Parkinson disease, neuromuscular diseases, muscular dystrophy, dysautonomias, peripheral nerve injuries, SCI, myelomeningocele, and other neurologic disorders have been shown to be difficult, challenging to manage, and often the primary disabling and handicapping feature for patients.^{4,90} Solving bowel problems significantly enhances quality of life.^{4,90}

NBD results from autonomic and somatic denervation and commonly produces FI, constipation, and difficulty with evacuation (DWE). The prevalence of FI and fecal impaction ranges from 0.3% to 5.0% in the general population. The prevalence of DWE ranges from 10% to 50% among the hospitalized or institutionalized elderly.¹³⁵ Although many GI disorders can contribute to FI or DWE, disorders that impair the extrinsic (sympathetic, parasympathetic, or somatic) nervous control of the bowel and anorectal mechanisms are more common among the patient populations seen by psychiatrists.

Impact

Living with nausea, vomiting, bloating, abdominal pain, constipation, diarrhea, or FI profoundly impacts quality of life. It significantly affects nutrition, overall health, and sense of well-being. Loss of voluntary control over bowel function alters every facet of a person's life at home, at work, and in the community. The ability to spontaneously regulate and direct bowel function is essential for participation in everyday activities at home, work, or school. It plays a major role in one's ability to create and maintain relationships. It has a significant impact on such activities as recreation and travel. Anticipation, experience and memory of episodes of extremes of NBD can create considerable psychological, social, and emotional trauma.

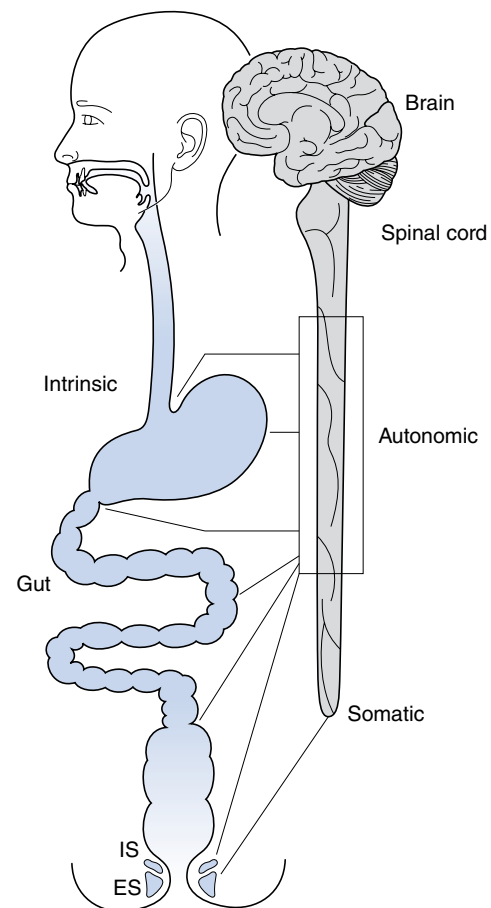
Incontinence, including of stool, has been repeatedly identified as a major factor for institutionalization. FI decreases return-to-home rates for stroke patients. Nursing home costs are higher for patients with FI.¹³⁵ Almost one-third of persons with SCI report or exhibit worsening of bowel function 5 years beyond their injury with 33% developing megacolon, suggesting inadequate long-term management.^{53,132} Of note, evidence has shown some improvement in outcomes for SCI bowel management.⁷⁰ When restoring normal defecation is not possible, *social continence* becomes the goal. Social continence is defined as predictable, scheduled, adequate defecations without incontinence at other times. It is often achievable by persons with NBD. Embarrassment and humiliation from FI frequently result in vocational and social disability, which adds substantial costs to the care related to NBD.

Neuroanatomy and Physiology of the Gastrointestinal Tract

In the past two decades, there has been significant advancement in the science of neurogastroenterology. This includes new discoveries in the basic physiology of the GI tract and interactions with the brain and spinal cord, autonomic nervous system and enteric nervous system (ENS), and somatomotor system (pharyngeal muscles and swallowing, pelvic floor muscles and defecation, continence and pelvic pain). Normal functioning of the stomach and intestines requires coordination of muscle contraction, digestion and absorption of nutrients, and regulation of blood flow. The neural schema of the GI tract is much more complex than initially thought. Neural control of the GI tract is an extremely organized and integrated hierarchy of mechanisms that involve the central nervous system (CNS; brain, spinal cord), autonomic nervous system (sympathetic and parasympathetic), and the ENS (Figs. 22.1 and 22.2).^{151,152}

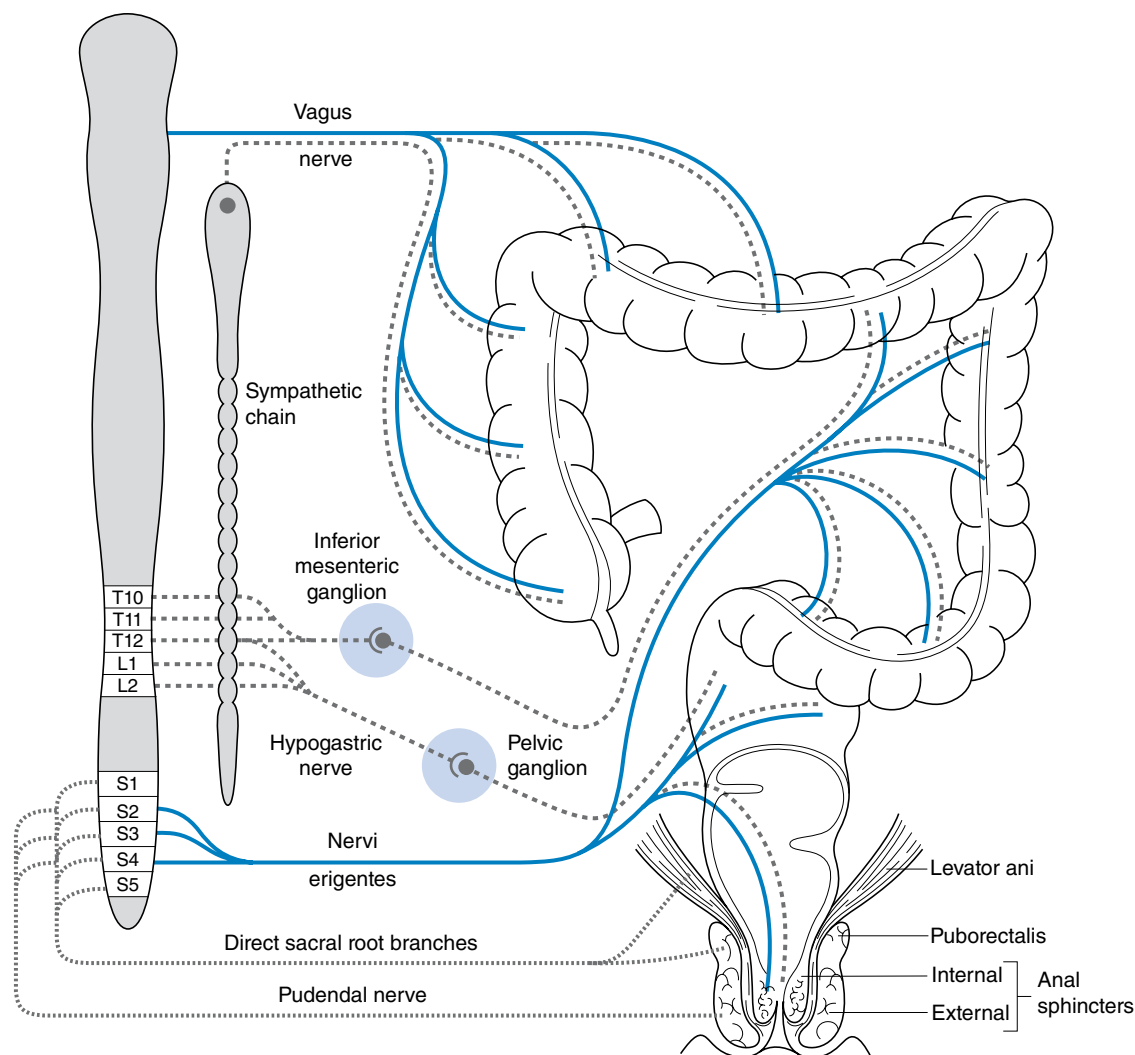
Enteric Nervous System

The ENS is a distinct network that has its own set of neurons that coordinate sensory and motor functions. In the ENS, the



• **Fig. 22.1** Gastrointestinal function is coordinated by three nervous systems: somatic, autonomic, and the enteric nervous system intrinsic to the gut wall. ES, External sphincter; IS, internal sphincter. (Courtesy John C. King, MD and University of Texas Health Science Center, San Antonio.)

^aReferences 1,31,66,122,129,145.



• **Fig. 22.2** Neurologic levels and pathways for the sympathetic, parasympathetic, and somatic nervous system innervation of the colon and anorectum. Not shown is the enteric nervous system, which travels along the bowel wall from esophagus to internal anal sphincter and forms the final common pathway to control the bowel wall smooth muscle. (Courtesy John C. King, MD and University of Texas Health Science Center, San Antonio.)

ganglia are interconnected, which allows for integration and processing of data (as opposed to autonomic ganglia, which only serve as relay centers for stimuli transmitted from the CNS). There are three different types of neurons in the ENS, which are sensory neurons, interneurons, and motor neurons.^{151,152} Sensory neurons are triggered by thermal, chemical, or mechanical stimuli and transform these sensations into action potentials that are conducted into the gut wall network. Interneurons serve as conduits between the sensory and motor neurons. The numerous synapses between interneurons create a highly organized circuitry that processes sensory input from the gut and other parts of the nervous system and integrates and generates reflex responses to these stimuli. Motor neurons are the final common pathway. They receive and translate signals to the gut (mucosa, muscle, vasculature) that affect digestive, interdigestive (i.e., between meals), and emetic functions based on the transmitters released.^{151,152}

It is now known that there is automatic feedback control in the ENS, with the neurons being in close proximity to the stomach and intestines. This can be manifested as reflex circuits that systematize reflex responses to sensory signals, integrative circuits that coordinate motor patterns (migrating motor complex [MMC], digestive activity, giant migratory contractions [GMCs]),^{151,152} or as a pattern generating activity that is generated when a “command neuron” is actuated and a rhythmic, repetitive behavior occurs.¹⁵² The ENS is the key to proper functioning of the entire GI tract. This collection of highly organized neurons is situated in two primary layers: the submucosal (Meissner) plexus and the intramuscular myenteric (Auerbach) plexus. Combined, these plexi interconnect an estimated 10 to 100 million neurons, plus two to three glial cells per neuron. The ENS glial cells resemble CNS astrocytes and are much less abundant than the 20 to 50 glial cells per neuron in the CNS.⁴⁵ The coordination of segment-to-segment function

is largely regulated by the ENS.¹⁴⁸ The ENS also has its own blood-nerve barrier, similar to the blood-brain barrier of the CNS.¹⁰⁸

Enteric Nervous System: Relationships to the Spinal Cord and Brain

Sensory information from *vagal afferents* in the ENS is relayed to the nodose ganglia (caudal ganglion of the vagus) and consequently to the nucleus tractus solitarius (NTS) and area postrema in the medullary area of brainstem. The NTS and the area postrema send signals to the rostral centers in the brain.^{151,152} The brain processes the data and projects descending connections to the dorsal vagal complex. The brain also participates in vagovagal reflex circuits, which continuously monitor and promptly modify responses to changes in the chemical, thermal, and mechanical environment in the whole gut (mostly in the esophagus, stomach, duodenum, gall bladder, and pancreas). Sensory stimulation through this vagal circuit does not appear to reach the level of consciousness.^{151,152}

Besides making connections with the NTS and the area postrema in the brainstem, vagal afferents synapse in the dorsal motor nucleus of the vagus (DVN) and the nucleus ambiguus, creating the dorsal vagal complex. The incoming sensory information is integrated and shared with the forebrain and brainstem by the dorsal vagal complex. Brain coordination and influences (conscious and nonconscious) are translated to the dorsal vagal complex, where the DVN and nucleus ambiguus represent the efferent arm of the reflex pathway. It is the final common pathway from the brain responsible for precise control of muscle, glandular, and circulatory responses of the GI tract.^{149,151,152} Multiple neurotransmitters are involved in the intricate conduction of impulses among the neuronal circuits in the dorsal vagal complex. Approximately 30 are identified and consist of acetylcholine, biogenic amines, amino acids, nitric oxide, peptides, etc.^{151,152}

Spinal afferents (splanchnic and pelvic) send sensory impulses through the dorsal root ganglia (or prevertebral sympathetic ganglia), which are then conducted to the dorsal horn (laminae I, II, V, X) of the spinal cord and the dorsal column nuclei.^{47,102} The dorsal column is thought to play a greater role in nociceptive transmission of messages from the gut than the spinothalamic or spino-reticular tracts. The conscious perception of pain in the digestive tract is largely mediated through spinal afferents. The spinal cord modulates the conduction of neural messages (both nociceptive and nonnociceptive) to the higher brain.⁴⁷ Other somatic and visceral sensory stimuli from the vagina, uterus, bladder, colon, and rectum are conveyed through the dorsal horn and dorsal column in the spinal cord as well. Likewise, somatic afferents supply sensory input from the muscles of the pelvic floor through the pudendal nerve to the sacral region of the spinal cord.^{10,47,101,102}

Brain centers modulate the nociceptive impulses from the bowel that are relayed to the dorsal horn of the spinal cord. These descending pathways are facilitative or inhibitory or both; based on the visceral stimulus; and can alter the perception of pain in the digestive tract.^{47,119} The neurotransmitters serotonin, noradrenalin, and dopamine are released by these descending pathways as they synapse within the spinal cord.¹⁵² Data from the brain and descending vagal pathways are conducted to the spinal cord and then to the preganglionic neurons in the thoracolumbar area (modulates the sympathetic response) and the sacral area (modulates the parasympathetic response). Sympathetic or

parasympathetic output is translated to the neurons in the ENS circuitry that can be excitatory or inhibitory to motor neurons, gastric or digestive glands, and secretory mechanisms.^{47,152}

The Alimentary Neuromotor System

The muscles of the GI tract carry out essential functions throughout the gut that include propulsion, grinding, mixing, absorption, storage, and disposal. The submucosal gut wall is composed of “self-excitabile” smooth muscles that contract in an all-or-none manner. It spontaneously responds to stretch and can be independent of neural or endocrine control. The interstitial cells of Cajal act like a pacemaker and allow propagation of electrical slow waves into the circular muscle layer which generates spontaneous muscle contraction. Depolarization occurs in an electrical syncytium where action potentials are conducted in three dimensions from one smooth muscle fiber to another via gap junctions.^{47,152}

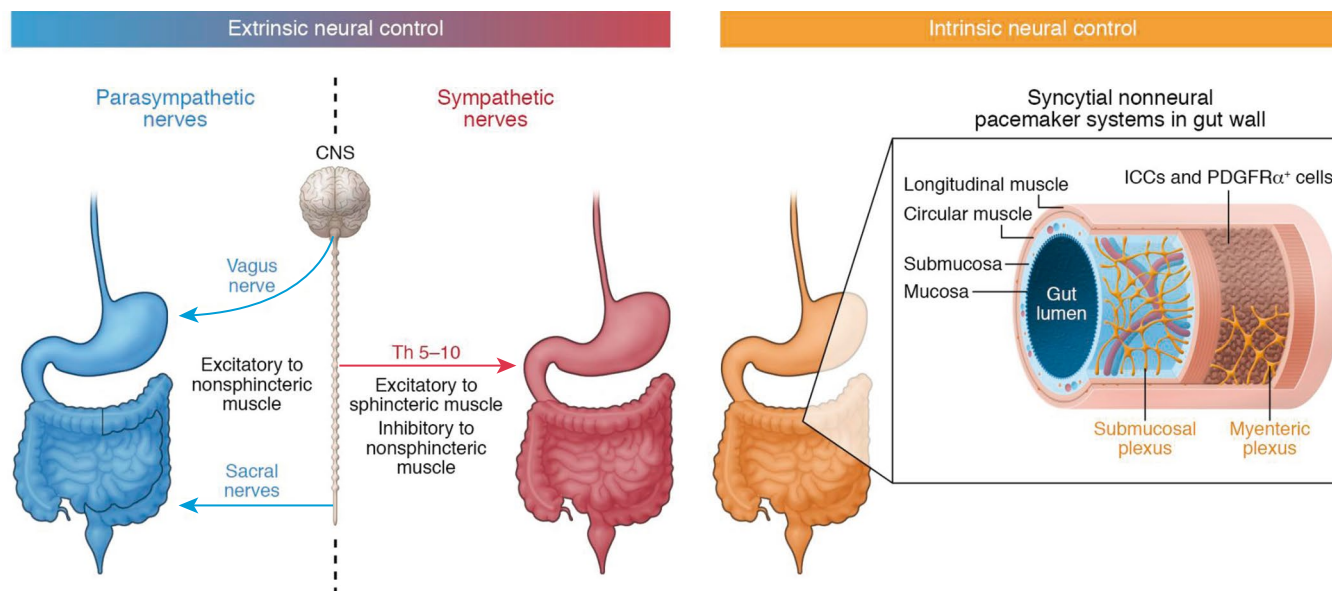
The sympathetic and parasympathetic nervous systems modulate the ENS, rather than directly controlling the smooth muscles of the bowel¹⁴⁸ (Fig. 22.3). The smooth muscles of the bowel also have their own electromechanical automaticity, which is directly modulated by the inhibitory control of the ENS.^{45,108} Sympathetic nervous system stimulation tends to promote the storage function by enhancing anal tone and inhibiting colonic contractions, although little perceptible clinical deficit occurs from bilateral sympathectomy.⁷²

Parasympathetic activity enhances colonic motility, and its loss is often associated with DWE, including impactions and functional obstructions (e.g., Ogilvie pseudo-obstructive syndrome).⁷²

The ENS and sympathetic postganglionic neurons transmit excitatory or inhibitory messages to secretomotor neurons. These secretomotor neurons release acetylcholine and vasoactive intestinal polypeptide when there is excitatory activation from paracrine stimulation by mucosal and submucosal cells such as enterochromaffin cells, mast cells, and other immune and inflammatory cells. Acetylcholine and vasoactive intestinal polypeptide are released at the neuroepithelial and neurovascular junctions. This promotes gut secretion of water, sodium chloride, bicarbonate, and mucus drawn from the intestinal glands. In addition, dilatation and increase in blood flow occur with release of nitric oxide from vascular endothelium.^{6,13,47} Inhibitory influence reduces neuronal firing from secretomotor neurons by hyperpolarization of membranes. Sympathetic release of norepinephrine from nerve endings of the α_2 -noradrenergic receptors inhibit secretomotor actuation, preventing the release of excitatory neurotransmitters. As a result, there is a decrease in secretion of water and electrolytes into the lumen and a congruent shunting of blood from the splanchnic to systemic circulation.^{47,88,104}

Gastric Motility

Based on its motility pattern the stomach is divided into an upper and lower portion. The upper portion (fundus) has sustained, low frequency contractions and has a tonic pattern. The lower portion (antrum) has intermittent, powerful contractions and has a phasic pattern. The fundus acts as a reservoir and accommodates incoming food, which inhibits contraction and allows the stomach to stretch without significant increase in pressure. The antrum is a mixer that generates propulsive waves that accelerate as food is propagated toward the pylorus. The amount and consistency of food in the fundus regulates excitatory and inhibitory influence and adjustments to volume and pressure.^{47,71}



• **Fig. 22.3.** Control of gut motility: extrinsic autonomic neural control, enteric nervous system, and smooth muscle function. The *left panel* shows extrinsic neural control, including the vagus and sacral parasympathetic nerves (blue) and the sympathetic innervation (red) originating from levels thoracic 5 to lumbar 2 in the spinal cord. The *right panel* shows the organization of the enteric nervous system, which demonstrates plexuses in the submucosal and intermuscular layers called the submucosal plexus and myenteric plexus. In addition, there are pacemakers in the wall of the intestine, including the interstitial cells of Cajal (ICCs) and fibroblast-like cells bearing receptors for platelet-derived growth factor receptor alpha $PDGFR\alpha$, which convey the neural stimulus to the smooth muscle cells. These pacemaker cells and plexuses help to coordinate muscular and secretory functions of the digestive tract. (From Camilleri M. Gastrointestinal motility disorders in neurologic disease. *J Clin Invest.* 2021;131(4):e143771. Fig 1.)

Intestinal Motility

The ENS is designed to control the various patterns of motility in the intestinal tract. The interdigestive MMC pattern occurs during fasting in the stomach and the small intestine.^{47,147} It seems to be influenced by the hormone motilin and is responsible for removal of waste from the intestinal lumen throughout the fasting period.^{47,89} When a meal is ingested, the postprandial segmentation (“mixing”) pattern of motility commences as digestion transpires. The brainstem sends signals that are transmitted to vagal efferents, which convert MMC motility to segmentation motility with the increase in bulk and nutrients, especially lipids (medium-chain triglycerides). This subsequently becomes peristaltic motility, which is propagated through brief segments of intestine at a time.^{11,47} Peristaltic activity gradually develops into powerful contractions sustained through long portions of circular muscle along the small and large bowel. These GMCs propel waste through the lumen, particularly in the large intestine.^{47,64,123}

Motility of the Anus, Rectum, and Pelvic Floor

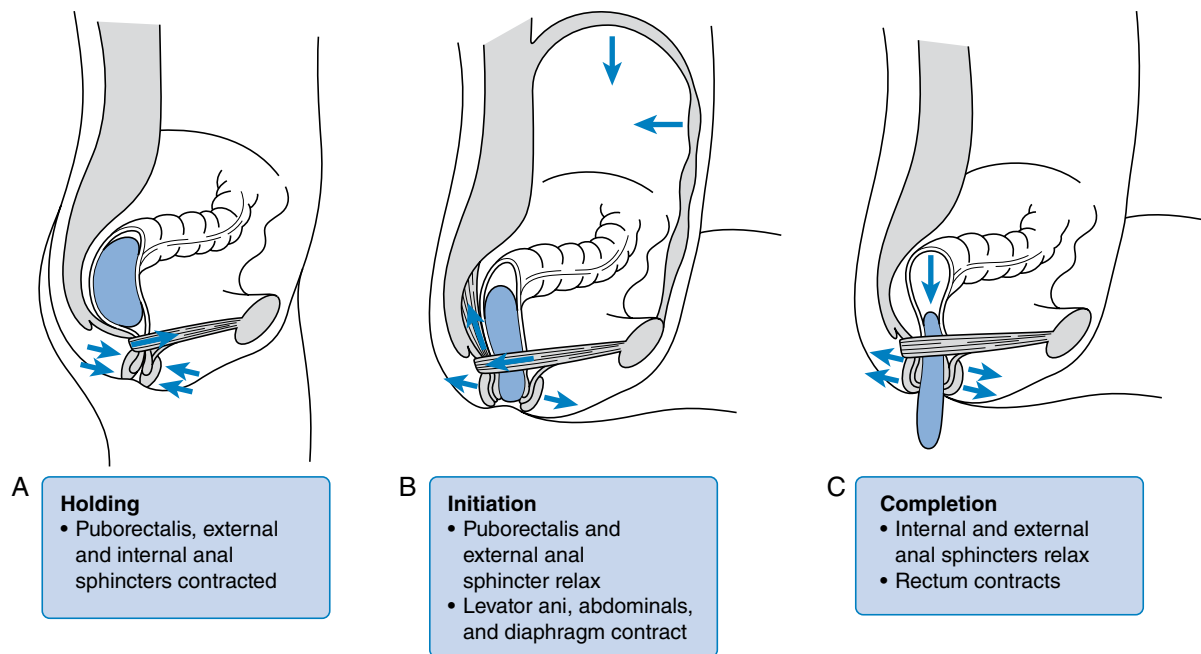
Normal defecation and maintenance of fecal continence entail a highly coordinated mechanism that involves the levator ani, puborectalis, the external anal sphincter (EAS), and internal anal sphincter (IAS) muscles. The pelvic floor includes the levator ani and the underlying sheets of muscle, which form a sling around the distal rectum. The levator ani, puborectalis, and EAS are skeletal muscles that constantly maintain tone supporting pelvic organs in place against the forces of gravity and maintain fecal

continence by kinking the rectum toward the pubis.^{48,58} Simultaneous contraction of these muscles prevent the involuntary loss of stool and help to maintain the regular pattern of defecation.^{40,47}

Physiology of Normal Defecation

The colon is a reservoir for food waste until it is convenient for elimination. It also acts as a storage device as long as the colonic pressure is less than that of the anal sphincter mechanism. Fecal elimination occurs when colonic pressure exceeds that of the anal sphincter mechanism. Another function of the colon is to reabsorb fluids, with up to 30 L/day reabsorbed through the large and small bowel walls and typically only 100 mL of water lost in feces. The colon also reabsorbs gases, with 90% of the 7 to 10 mL of gases produced by intracolonic fermentation daily absorbed rather than expelled. The colon additionally provides an environment for the growth of bacteria needed to assist in digestion and serves to absorb certain bacterial breakdown products as well.⁵⁰

The rectum is usually empty until just prior to defecation. Perception of external rectal contents and pressures¹²⁷ is essential for signaling voluntary contraction of the anal sphincter. Normal defecation begins with reflexes triggered by rectosigmoid distention produced by approximately 200 mL of feces (Fig. 22.4).¹¹⁰ A rectorectal reflex occurs in which the bowel proximal to the distending bolus contracts and the bowel wall distally relaxes, serving to propel the bolus further downward. Reflex relaxation of the internal sphincter and stretching of the puborectalis muscle also occurs, which is enhanced by, but does not require, an extrinsic nerve supply. This relaxation, called the rectoanal inhibitory reflex, correlates



• **Fig. 22.4** (A) Defecation is prevented by a statically increased tone of the internal anal sphincter (IAS) and puborectalis, as well as by the mechanical effects of the acute anorectal angle. Dynamic responses of the external anal sphincter (EAS) and puborectalis to rectal distention reflexes or increased intraabdominal pressures further impede defecation. (B) To initiate defecation, the puborectalis muscle and EAS relax while intraabdominal pressure is increased by Valsalva maneuver, which is facilitated by squatting. The levator ani helps to reduce the acute anorectal angle to open the distal anal canal to receive the stool bolus. (C) Intrarectal reflexes result in continued IAS relaxation and rectal propulsive contractions, which help to expel the bolus through the open canal. (After Schiller LR. Fecal incontinence. In Sleisenger MH, Fordtran JS (eds). *Gastrointestinal Disease*. Philadelphia: WB Saunders, 1993, pp 934–953.)

with the urge labeled “the call to stool.”¹⁴⁸ One can then volitionally contract the levator ani to open the proximal anal canal and relax the external sphincter and puborectalis muscles. This allows a straighter, shorter, and more open anorectal passage (see Fig. 22.4), which permits the bolus to pass. Increasing the intraabdominal pressure by squatting and by a Valsalva maneuver assists bolus elimination. For 90% of normal individuals only the contents of the rectum are expelled, whereas 10% will clear the entire contents of the left or sigmoid colon from the splenic flexure distally.¹⁰⁵

However, one can elect to defer defecation by volitionally contracting the puborectalis muscle and EAS. The reflexive IAS relaxation subsequently fades, usually within 15 seconds, and the urge resolves until the IAS relaxation is again triggered. The rectal wall accommodates to the bolus by decreasing its wall tension with time, resulting in less sensory input and less reflex triggering from that particular bolus. This continence and reflex process is somewhat analogous to the function of the striated external urethral sphincter in volitional control of urinary voiding (see Chapter 21).

The external sphincter generally tenses in response to small rectal distentions via a spinal reflex, although reflexive relaxation of the external sphincter occurs in the presence of greater distentions.¹²⁷ These spinal cord reflexes are centered in the conus medullaris and are augmented and modulated by higher cortical influences. When cortical control is disrupted, as by SCI, the external sphincter reflexes usually persist and allow spontaneous defecation. During sleep, the colonic activity, anal tone, and protective responses to abdominal pressure elevations are all decreased, while rectal tone increases.^{108,148}

The “gastrocolonic response” or “gastrocolic reflex” refers to the increased colonic activity (GMCs and mass movements) in the first 30 to 60 minutes after a meal. This increased colonic activity appears to be modulated both by hormonal effects from release of peptides from the upper GI tract (gastrin, motilin, and cholecystokinin, which increase contractility of colonic smooth musculature) and by a reduction in the threshold for spinal cord-mediated vesicovescical reflexes.¹⁰⁸ Upper GI receptor stimulation also results in increased activity in the colon, possibly due to reflexively increased parasympathetic efferent activity to the colon. The possibility of a purely ENS-mediated activation exists, although the small bowel and colon motor activities do not seem to be synchronized. In persons with SCI, the measured increase in colonic activity after a meal is blunted as compared with normals.¹⁰⁸ The gastrocolonic response is often used therapeutically, even in SCI patients, to enhance bowel evacuation during this 30- to 60-minute postprandial time frame.¹ Occasionally certain foods can serve as trigger foods that are especially likely to induce bowel evacuation shortly after consumption.

The resting anal canal pressure is largely determined by the angulation and pressure at the anorectal junction by the puborectalis sling and smooth muscle internal sphincter tone. Continence is maintained by the anal sphincter mechanism,^{94,129} which consists of the IAS, EAS, and the puborectalis muscle.⁹⁴ Only approximately 20% of the anal canal pressure is due to the static contraction of the somatically innervated striated EAS.⁹ The EAS and puborectalis muscle are the only striated skeletal muscles whose normal resting state is tonic contraction. These muscles

consist mainly of slow-twitch fatigue-resistant type I fibers (unlike the situation in nonupright animals such as the cat or dog in which it consists of predominately type II fibers).⁹ Anal pressure can be increased volitionally by contracting the EAS and puborectalis muscles. However, maximum volitional squeeze pressures are not as high as can be generated reflexively against Valsalva pressure. The EAS is physically larger than the internal sphincter, and its contraction is under both reflex and volitional control. The volitional control is learned during the course of normal maturation. Normal baseline reflex action of the anorectal mechanism allows spontaneous stool elimination.¹²⁵ The EAS is innervated by the S2 through S4 nerve roots via the pudendal nerve. The puborectalis muscle is innervated by direct branches from the S1 to S5 roots (see Fig. 22.1).¹⁰⁶ The remarkable degree of learned EAS coordination and discriminating rectal sensation allows the selective discrete passage of gas while juggling a variable mixture of solids, liquids, and gases.

Pathophysiology of Gastrointestinal Dysfunction

A whole range of neurologic diseases affecting central, peripheral, and intrinsic enteric nervous innervation has been demonstrated to result in disorders in various segments of the bowel. These are predominantly characterized by disturbances in gastroesophageal, small or large intestinal motility, and sensation. This chapter concentrates on pathologic conditions pertaining to extrinsic neural regulation (neurological disorders, degenerative conditions of the autonomic nervous system, lesions of the spinal cord, and peripheral neuropathies) as well as muscular disorders that influence gastrointestinal motility. Symptoms of dysphagia, vomiting, bloating, abdominal discomfort and pain, constipation, and incontinence have been described in individuals with neurologic ailments.^{4,18,19,107} Exact and detailed pathophysiologic mechanisms of GI dysfunction in the various neurologic diseases are not very well understood. Recent studies, however, have been able to identify various mechanisms of symptom production that might facilitate therapeutic strategies.

The Rome IV classification is a diagnostic system developed to standardize the diagnosis and classification of functional gastrointestinal disorders (FGIDs), also known as disorders of gut-brain interaction. Updated in 2016, Rome IV criteria, reflect advances in understanding the pathophysiology, diagnosis, and treatment of FGIDs. It can be helpful in classifying chronic or recurrent symptoms that arise from neurogenic gastrointestinal dysfunction that cannot be explained by structural or biochemical abnormalities.³⁴

Nausea, Vomiting, Bloating, Early Satiety

The syndrome of nausea, vomiting, bloating, and early satiety in the setting of neurologic conditions without mechanical obstruction can herald motility problems in the GI tract. Neurologic dysfunction that affects the inhibitory motor neurons in the ENS at any level of the neural axis from the brain, spinal cord, and afferent and efferent nerves can lead to dyscoordinated overactivity of the gastric or intestinal/colonic musculature. When inhibitory motor neurons are inactivated or destroyed by disease, there is a continuous, nonsystematic contraction of the circular muscles incapable of forward propulsion, causing functional obstruction.^{150,151} This can be manifested as dysphagia, gastroparesis, or chronic intestinal/colonic pseudo-obstruction, which might be

associated with anorexia, abdominal pain, diarrhea, and constipation⁵⁷. Inhibitory motor neurons can be affected by denervation of central connections autonomic neuropathy, by dysfunction of neurons in the myenteric plexus, or from degeneration of smooth muscle.^{6,7}

Abdominal Pain and Discomfort (Box 22.1)

Abdominal pain and discomfort can arise from GI tract distention and powerful contractions. High-threshold and silent mechanoreceptors sense severe distention and intense contractions when there is ischemia, injury, inflammation, or obstruction. Mechanical and chemical irritants stimulate mechanoreceptors in the ENS and translate signals to the brain and spinal cord from muscle stretching and contractions.^{101,150} Of note, in individuals with thoracic SCIs at the 6th thoracic level and above, this can result in vague nonlocalized discomfort or autonomic dysreflexia. The presence and persistence of ischemia, injury, and inflammation can elicit abdominal pain from a number of mechanisms: Spinal afferent nerves express receptors for inflammatory mediators, and the release of bradykinin, adenosine triphosphate, adenosine, prostaglandins, leukotrienes, histamine, and mast cell proteases heightens sensitivity of these spinal sensory endings.¹⁵⁰ Amplified levels of 5-HT₃ released from hyperactive 5-HT₃ receptors at the vagal and spinal nerve endings facilitate production of pain. Excessive induction of serotonergic receptors by 5-HT₃ is due to a defective serotonin transporter.^{59,75,150} Central sensitization occurs in the spinal cord from constant stimulation of the C fibers in the dorsal horn and dorsal column nuclei, and neurons become hypersensitive and persistently activated. The “wind up” phenomenon develops with the actuation of NMDA receptors from the release of glutamate from C fibers. The reactive state of dorsal horn and dorsal column neurons becomes a memory imprint in the spinal cord.^{2,3,150} Mechanoreceptors in the gut wall can receive a deluge of signals from mechanical irritation. This barrage of impulses conducted to the brain can be interpreted as nociceptive because of the overwhelming stimulation.^{150,152} The descending facilitative modulation of the brain through the dorsal horn can promote recognition of ordinary sensory input as innocuous. The brain misreads normal nonconscious signals relayed from mechanoreceptors as noxious.^{119,150}

^bReferences 7,28,56,57,128,142,150.

• Box 22.1 Rome IV Classification for Functional Abdominal Bloating/Distension (C4)

Diagnostic criteria*

Must include both of the following:

1. Recurrent bloating and/or distension occurring on average at least 1 day/week; abdominal bloating and/or distension predominates over other symptoms.**
2. There are insufficient criteria for a diagnosis of irritable bowel syndrome, functional constipation, functional diarrhea, or post-prandial distress syndrome.

*Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

**Mild pain related to bloating may be present as well as minor bowel movement abnormalities
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Diarrhea (Box 22.2)

Neurologic dysfunction can present with frequent passage of watery stools. Diarrhea occurs when there is overstimulation of secretomotor neurons by histamine from inflammatory and immune-mediated cells in the mucosa and submucosa and/or vasoactive intestinal peptide and serotonin from mucosal enterochromaffin cells. Moreover, these substances influence presynaptic inhibitory receptors to block the release of norepinephrine from the postganglionic sympathetic fibers that inhibit secretomotor neurons.^{149,150}

Pelvic and Anorectal Pain (Box 22.3)

Acute and chronic pelvic/anorectal pain can be a feature of neurologic disorders. This may present as nociceptive pain that arises

• BOX 22.2 Rome IV Criteria for Functional Diarrhea (C3)

Diagnostic criterion*

Loose or watery stools, without predominant abdominal pain or bothersome bloating, occurring in more than 25% of stools.**

IBS, Irritable bowel syndrome.

*Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

**Patients meeting criteria for IBS-D (diarrhea-predominant IBS) should be excluded.

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• BOX 22.3 Rome IV Classification for Functional Anorectal Pain (F2)

F2a. Levator Ani Syndrome

Diagnostic criteria*

Must include **all** of the following:

1. Chronic or recurrent rectal pain or aching
2. Episodes last 30 minutes or longer
3. Tenderness during traction on the puborectalis
4. Exclusion of other causes of rectal pain such as inflammatory bowel disease, intramuscular abscess, anal fissure, thrombosed hemorrhoids, prostatitis, coccygodynia and major structural alterations of the pelvic floor

*Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

F2b. Unspecified Functional Anorectal Pain

Diagnostic criterion*

Symptom criteria for chronic levator ani syndrome but no tenderness during posterior traction on the puborectalis muscle

*Criterion fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

F2c. Proctalgia Fugax

Diagnostic criteria*

Must include **all** of the following:

1. Recurrent episodes of pain localized to the rectum and unrelated to defecation
2. Episodes last from seconds to minutes with a maximum duration of 30 minutes
3. There is no anorectal pain between episodes
4. Exclusion of other causes of rectal pain such as inflammatory bowel disease, intramuscular abscess, anal fissure, thrombosed hemorrhoids, prostatitis, coccygodynia and major structural alterations of the pelvic floor

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from damage to sites sensitive to pain and classified as somatic or visceral in nature, and/or nonnociceptive pain, classified as neuropathic or psychogenic. Pelvic/anorectal pain can be caused by any anatomical structure in the abdominal and pelvic regions. The effects of the gastrointestinal system, blood vessels, muscles and fascia of the abdominal wall and pelvic floor, the ureters and bladder, and the peripheral and central nervous systems must therefore be taken into account in addition to the upper reproductive organs.⁴⁹ Many individuals with neurologic disorders like SCI, MS, stroke, etc., which can result in pelvic floor dysfunction, associated with neurogenic bladder, bowel, and sexual dysfunction and have been reported to develop pelvic/anorectal pain. This demonstrates how the reproductive, urinary tract, and colon systems are intimately related.^{51,99}

Defecatory Dysfunction: Constipation and Fecal Incontinence (Box 22.4)

Neurologic compromise can result in slow transit constipation and functional defecation disorders. Slow transit constipation is described as a deficiency in the propulsive forces within the colon, occurring in the absence of dyssynergic defecation. Functional defecation disorders encompass dyssynergic defecation, characterized by the paradoxical contraction or insufficient relaxation of the pelvic floor musculature during the effort to defecate.¹¹⁵

Constipation (Box 22.5)

Constipation can be an enigma in neurologic states. Constipation can usually be defined as fewer than three bowel movements per week. However, constipation is not strictly limited to this definition because it can include symptoms of abdominal discomfort, bloating, distention, increased straining, sensation of incomplete emptying and anal blockage, and need for manual removal of stool.¹² Constipation related to NBD is often due a number of factors that are caused directly by the neurologic dysfunction and other indirect factors, such as decreased mobility, altered appetite, altered fluid intake, changes to lifestyle routine, medication effects, and psychological issues.

Constipation effects of NBD results from impaired or absent muscle and sensory function that affects defecation. In supraconal lesions (above lumbar level 1), rectal emptying becomes inadequate due to poor propulsive forces and/or increased resistance. This is typically due to both high anal resting pressures and rectosphincteric dyssynergia with paradoxical contraction or incomplete pelvic floor relaxation during defecation.^{77,137,139} Infraconal lesions (at lumbar level 1 and below) will present with the opposite picture: weak or absent sphincter tone, and atonic pelvic floor muscles. Both of these conditions result in obstructed defecation and are not uncommonly associated with prolonged colonic transit.^{77,137,139}

Infrequent, incomplete emptying of hard stools is a result of a decrease in secretion of water and electrolytes into the lumen due to reduced excitation of the secretomotor neurons in the ENS or prolonged stool transit time and overabsorption of water. Norepinephrine released by sympathetic stimulation inhibits the firing of secretomotor neurons by hyperpolarization. Release of excitatory neurotransmitters is reduced in the secretory epithelium, decreasing the secretion of water and electrolytes.^{150,152} Lack of rectal sensation and decreased urge to defecate is associated with constipation in various conditions that present with lesions of the brain, spinal cord, sacral nerves, and hypogastric and pudendal nerves.

• BOX 22.4 Rome IV Criteria for Functional Defecation Disorders (F3)

Diagnostic criteria*

Must include **all** of the following:

1. The patient must satisfy diagnostic criteria for functional constipation and/or irritable bowel syndrome with constipation
2. During repeated attempts to defecate, there must be features of impaired evacuation, as demonstrated by 2 of the following 3 tests:
3. Abnormal balloon expulsion test
4. Abnormal anorectal evacuation pattern with manometry or anal surface EMG
5. Impaired rectal evacuation by imaging

*Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

Subcategories F3a and F3b apply to patients who satisfy criteria for FDD

F3a. Inadequate Defecatory Propulsion

Diagnostic criterion*

Inadequate propulsive forces as measured with manometry with or without inappropriate contraction of the anal sphincter and/or pelvic floor muscles**

*Criterion fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

**This criterion is defined by age- and gender-appropriate normal values for the technique

F3b. Dyssynergic Defecation

Diagnostic criterion*

Inappropriate contraction of the pelvic floor as measured with anal surface EMG or manometry with adequate propulsive forces during attempted defecation**

EMG, Electromyography; FDD, functional defecation disorder.

*Criterion fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

**This criterion is defined by age- and gender-appropriate normal values for the technique

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• BOX 22.5 Rome IV Criteria for Functional Constipation (C2)

Diagnostic criteria*

1. Must include **two or more** of the following:**
2. Straining during more than ¼ (25%) of defecations
3. Lumpy or hard stools (Bristol Stool Form Scale 1-2) more than ¼ (25%) of defecations
4. Sensation of incomplete evacuation more than ¼ (25%) of defecations
5. Sensation of anorectal obstruction/blockage more than ¼ (25%) of defecations
6. Manual maneuvers to facilitate more than ¼ (25%) of defecations (e.g., digital evacuation, support of the pelvic floor)
7. Fewer than three SBM per week
8. Loose stools are rarely present without the use of laxatives
9. Insufficient criteria for irritable bowel syndrome

*Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

**For research studies, patients meeting criteria for opioid-induced constipation (OIC) should not be given a diagnosis of FC because it is difficult to distinguish between opioid side effects and other causes of constipation. However, clinicians recognize that these two conditions may overlap.

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Functional outlet obstruction can occur due to delayed colonic transit times and overdistention of the rectum due to the lack of perineal and rectoanal sensation.¹⁵⁰

Colonic motility and stool propulsion are known to be affected by SCI. A questionnaire method to study subjects with SCI levels above L2 found that 58% of subjects with chronic SCI

• BOX 22.6 Rome IV Criteria for Fecal Incontinence (F1)

Diagnostic criterion*

Recurrent uncontrolled passage of fecal material in an individual with a developmental age of at least 4 years

*Criterion fulfilled for the last 3 months. For research studies, consider onset of symptoms of at least 6 months previously with 2-4 episodes of FI over 4 weeks.

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had constipation (defined as two or less bowel movements per week or the requirement for digital evacuation).³¹ There have been several studies done on colonic motility using colon transit time using swallowed markers and serial radiographs.

Whole gut transit time and colon transit time of the ascending, transverse, and descending colon were significantly prolonged in subjects with acute supraconal or conal/cauda equina lesions, although rectosigmoid transit time was significantly prolonged only in acute conal/cauda equina lesions.⁸¹ Whole gut transit time and colon transit time were prolonged in subjects with chronic supraconal or conal/cauda equina lesions. Transverse and descending colon transit times were significantly prolonged in subjects with supraconal lesions, and transverse, descending colon and rectosigmoid colon transit times were significantly prolonged in conal/cauda equina lesions.⁸¹ A more recent study showed the same findings in chronic supraconal lesions.³⁵ Other studies have found colon transit delay in supraconal and infraconal lesions in complete and incomplete SCI.^{137,139}

A wireless motility capsule has been able to provide information on GI pH and pressures; gastric, small intestinal, and colonic segmental transit times; and total gut motility. The pill is ingested with a special meal, and information is transmitted to an external recorder worn over the abdomen. A study completed in people with SCI using the wireless motility capsule showed that gastric emptying, colon transit time, and whole gut transit time is delayed in people with paraplegia and tetraplegia, compared with controls.¹⁴⁶ A GI magnetic pill with a motility tracking system showed orocecal transit time was longer than normal both in individuals with supraconal lesions and in individuals with infraconal lesions. Individuals with lesions above T6 had slower gastric emptying than those with lesions below T6.⁴³ These systems provide the advantage of comprehensive GI transit testing with one procedure as opposed to traditional cumbersome testing, which requires multiple hours over multiple days.

Fecal Incontinence (Box 22.6)

Frequent and repetitive FI described as the unconscious loss of stool is more frequent in neurologic conditions with lesions affecting the lumbar spinal cord, cauda equina, S2-4 nerves, pudendal nerve, and pelvic floor nerves. Denervation leads to impaired perineal and rectoanal sensation, aberrant internal sphincter contraction, loss of tone, and weakening of the pelvic floor muscles and the EAS. This underlies unexpected loss of stool and abnormal defecation and diminished support for pelvic structures.^{26,47} Parasympathetic augmentation can occur and might further complicate matters because it contributes to weakness of the internal sphincter, stimulates peristalsis, and increases the risk for incontinence. FI also occurs with lesions above the lumbar level due to loss of voluntary control over defecation resulting in uncontrollable

reflex defecation and lack of sphincter pelvic floor muscle coordination. In contrast, thoracic and cervical lesions have hypertonicity of the sphincter and pelvic floor muscles and/or can present with anorectal sphincter dyssynergia, resulting from paradoxical contraction of the sphincter with pelvic floor relaxation.^{37,77,93,103,137} Furthermore, it is always important to rule out overflow incontinence from the “ball valve” effect of a partially blocking fecalith due initially to constipation.

Reflexic Supraconal Neurogenic Bowel Dysfunction

Any destructive CNS process above the conus, from SCI to dementia, can lead to supraconal NBD pattern. Spinal cortical sensory pathway deficits lead to decreased ability to sense the urge to defecate. However, most persons with SCI sense a vague discomfort when excessive rectal or colonic distention occurs. It has been reported that 43% have chronic complaints of vague abdominal distention discomfort that eases with bowel evacuation.^{92,132} These sensations might be mediated by autonomic nervous system afferent fibers bypassing the zone of SCI via the paraspinal sympathetic chain or by means of vagal parasympathetic afferents.

Anal sensation, rectal compliance, sphincter tone and function, and pelvic floor relaxation have been studied in SCI subjects using anorectal physiologic testing to assess defecation. Resting sphincter tone has been found to be normal or exaggerated compared with normal controls.^{80,93} Rectal sensation was impaired in complete SCI lesions and less impaired in incomplete lesions.^{93,138} Voluntary anal squeeze was present in most patients with motor

incomplete lesions and absent in all patients with motor complete lesions.¹³⁸ Rectal sphincter dyssynergia and the absence of sphincter and pelvic floor relaxation were observed during simulated defecation in supraconal lesions,^{31,93,138} most likely contributing to significant difficulty with defecation.¹³⁹ In addition, increased rectal tone and decreased rectal compliance⁹¹ with associated anal resistance with defecation and the attenuation of EAS contraction with increased intraabdominal pressure and during rectal distention also contribute to difficulty with defecation.^{80,93,139} This was demonstrated in both acute and chronic supraconal lesions.⁸⁰

SCI subjects with complete supraconal lesions were studied during the acute phase (5 to 21 days) after SCI. These same patients were reevaluated 6 to 14 months later. Total GI transit time was longer during the acute rather than the chronic phase. The supraconal neurogenic colon tended to have slower transit throughout the colon with less severe rectosigmoid dysfunction.⁸¹ Patients with supraconal NBD have spared reflex arc control of the rectosigmoid and pelvic floor. Internal sphincter relaxation upon rectal distention occurs in persons with SCI as well as in neurologically intact persons. Sufficient rectal distention might cause the external sphincter to completely relax, resulting in expulsion of the fecal bolus. Rectal sphincter dyssynergia does not necessarily correlate with bladder sphincter dyssynergia, but it often results in DWE.¹⁰⁶ The protective vesicorectal reflex, in which the external sphincter pressure increases in response to increased intraabdominal pressure, is usually intact (Table 22.1).⁹ Patients with supraconal NBD also have normal or increased anal sphincter tone, intact anocutaneous (or anal wink) and bulbocavernosus reflexes,¹²⁹ a palpable puborectalis muscle sling, and normal anal verge appearance (Fig. 22.5).

TABLE 22.1 Features of Colorectal Function in Normal Subjects and in Those With SUPRACONAL NBD, SUPRACONAL NBD With Posterior Rhizotomy, and INFRACONAL NBD

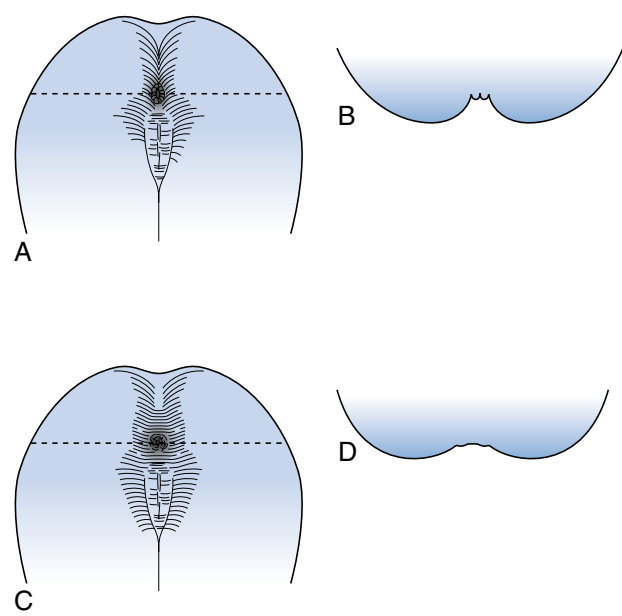
	Normal	SUPRACONAL NBD	SUPRACONAL NBD and Posterior Rhizotomy	INFRACONAL NBD
Bowel dysfunction	Normal colon activity and defecation	Chronic intractable constipation, fecal impaction, reflex defecation with or without incontinence	Chronic constipation, no defecation	Chronic constipation, fecal impaction, maximal in the rectum
Transit time (cecum to anus)	12–48 h	Prolonged >72 h	Very prolonged unless sacral nerve stimulator used	Prolonged >6 days, especially left colon
Colonic motility at rest	GMCs approximately 4 per 24 h	GMCs may be reduced in frequency	Reduced GMC	Reduced GMC
Colonic motility in facilitation response to stimuli	GMCs facilitated by defecation, exercise, and food ingestion	Less GMC facilitation by defecation, exercise, or food ingestion	Less GMC facilitation by defecation, exercise, or food ingestion	Less GMC facilitation by defecation, exercise, or food ingestion
Anal Sphincter Pressure (mm Hg)				
Resting tone	>30	>30	Normal	Reduced
Volitional squeeze	>30 (up to 1800)	Absent	Absent	Absent
Rectal compliance	Normal	Normal but sigmoid compliance decreased	Normal or increased	Rectum dilated, increased distension volume, increased compliance
Rectal Balloon Distension				
Effect on IAS	Normal rectoanal inhibitory reflex	Normal rectoanal inhibitory reflex	Normal rectoanal inhibitory reflex	Normal rectoanal inhibitory reflex

Continued

TABLE 22.1 Features of Colorectal Function in Normal Subjects and in Those With SUPRACONAL NBD, SUPRACONAL NBD With Posterior Rhizotomy, and INFRACONAL NBD—cont'd

	Normal	SUPRACONAL NBD	SUPRACONAL NBD and Posterior Rhizotomy	INFRACONAL NBD
Effect on EAS	Causes contraction	Causes contraction	No contraction	No contraction
Sensory perception threshold	<20 mL volume	None	None	None
Stimulation of rectal contraction	Induced by balloon distension	Giant rectal contractions stimulated readily	Rectal contraction stimulation	Rectal contraction stimulation
Vesicoanal reflex	Present (>50 mm Hg)	Present	Absent	Absent
Valsalva Protective Reflex				
Reflex defecation	Yes	Yes	Impaired	Impaired
Perianal sensation (cutaneous sensation of touch, pinprick)	Normal	No sensory perception	No sensory perception	Loss of perianal buttock sensation due to injury to sacral nerves
Anocutaneous reflex ("anal wink")	Present	Present; may be increased	Absent	Absent due to injury to afferent or efferent sacral pathways
Bulbocavernosus reflex	Present	Present; may be increased	Absent	Absent
Anal appearance	Normal	Normal	Normal	Flattened, scalloped, due to loss of EAS bulk

EAS, External anal sphincter; GMCs, giant migratory contractions; IAS, internal anal sphincter; NBD, neurogenic bowel dysfunction.
After Banwell JG, Creasey GH, Aggarwal AM, Mortimer JT. Management of the neurogenic bowel in patients with spinal cord injury. *Urol Clin North Am.* 1993;20(3):517–526; and Schiller LR. Fecal incontinence. In Sleisenger MH, Fordtran JS (eds). *Gastrointestinal Disease*. Philadelphia: WB Saunders, 1993, pp 934–953.



• **Fig. 22.5** Supraconal neurogenic bowel presents an appearance similar to normal (A, rear view; B, profile from above). Anal contour of the infraconal neurogenic bowel (C, rear view; D, profile from above), with its atrophic external anal sphincter, shows a flattened, scalloped-appearing anal area.

Hyporeflexic Infraconal Neurogenic Bowel Dysfunction

Polyneuropathy, conus medullaris or cauda equina lesions, pelvic surgery, vaginal delivery, or even chronic straining during defecation can impair the bilateral pelvic nerve somatic innervation of the anal sphincter mechanism. Anorectal physiologic studies show absence of anal resistance with defecation and absence of EAS contraction with intraabdominal pressure increased and during rectal distention.¹³⁹ Rectal tone and contractions are noted to be reduced or absent in the infraconal NBD.^{80,139} These conditions can also produce sympathetic and parasympathetic innervation deficits. If an isolated pudendal nerve insult has occurred, colonic transit times are normal and FI predominates. Distal colonic sluggishness can occur as a result of loss of parasympathetic supply. Segmental stool transit studies demonstrate prolonged transit through the rectosigmoid segments due to the lack of direct innervation from the conus.⁸¹ The addition of constipation and DWE to FI compounds difficulties. This is an especially problematic combination because the accumulation of a large amount of hard stool that can result from such colonic inertia can overstretch the weakened anal mechanism. This can result in a gaping, patulous, incompetent anal orifice, often with associated rectal prolapse. The denervation, atrophy, and overstretching of the EAS and IAS lead to loss of the protective IAS tone, which can result in stool soilage from the increased abdominal pressures associated with everyday

activities. Rectal distention leads to the expected internal sphincter relaxation, but attenuated or absent external sphincter protective contractions can result in FI or fecal smearing whenever boluses are present at the rectum. The presence of a large bolus in the rectal vault can further compromise the rectoanal angulation at the pelvic floor and contribute to paradoxical liquid incontinence around a low impaction, called the *ball-valve effect*.^{9,133,153} Patients with infraconal NBD may also have decreased anal tone due to the smooth muscle that makes up the internal sphincter. The anal-to-buttock contour typically appears flattened and “scalloped” (see Fig. 22.5) due to atrophy of the pudendal-innervated pelvic floor muscles and EAS.⁹ The anocutaneous reflex is absent or decreased (depending on the completeness of the lesion). The bulbocavernosus reflex is also weak if present (see Table 22.1). The anal canal is shortened (as compared with the normal 2.5- to 4.5-cm length), and the puborectalis muscle ridge might not be palpable. Excessive perineal descent and even rectal prolapse can occur with Valsalva maneuver. Interestingly, colonic transit time and whole gut transit time have been noted to be severely delayed in conal lesions as well.^{103,139}

Gastrointestinal Dysfunction in Common Neurologic Disorders

Brain Disorders

Strokes (ischemic or hemorrhagic), cerebral trauma, mass, infection, or other lesions cause a whole range of neurologic damage and complications to many areas of the brain. They commonly present with dysphagia, delays in gastric emptying and ileus (intestinal pseudo-obstruction), constipation, and FI. Brainstem or cranial nerve involvement often causes dysphagia. Studies report dysphagia occurs in approximately 45% of patients with stroke in the acute phase. Cerebral edema and high intracranial pressure play a role in GI ulceration, gastroparesis, or ileus through unknown mechanisms. Depending on the severity of the cerebral or brainstem involvement, dysphagia and ileus progressively improve. Approximately 23% of stroke patients were shown to have FI in stroke research, and constipation has been observed to be very common in clinical practice.^{4,19,91,107}

Parkinson Disease and Parkinson Plus Diseases

Parkinson and Parkinson-plus diseases (multiple system atrophy, progressive supranuclear palsy, corticobasal degeneration) are neurodegenerative diseases that affect various parts of the nervous system, including the cerebral cortex, basal ganglia, brainstem, cerebellum, and spinal cord. They are characterized by dementia, muscle rigidity, bradykinesia, resting tremor, dystonia, shuffling gait, and dysphagia. They also have dysautonomia, which is manifested as orthostatic hypotension, sexual dysfunction, gastric dysmotility, and constipation. Up to 50% of people with Parkinson disease have been found to have dysphagia and problems with defecation. In a study done on 98 people, 52.1% had dysphagia, 28.7% had constipation, and 65% had problems with defecation. These GI problems can be severe enough to compromise nutrition and cause severe morbidity.^{26,39,82,100,109,113}

Multiple Sclerosis

Multiple sclerosis is a demyelinating disease caused by an autoimmune process that involves different regions of the brain and

spinal cord. Initial presentations can be any neurologic sign or symptom, which can consist of paresis, spasticity, ataxia, paresthesias, visual problems, cognitive deficits, mood disorders, dysphagia, gut dysmotility, urinary incontinence or retention, FI, diarrhea, or constipation. In a study done on patients with multiple sclerosis, 68% were reported to have GI problems, with 43% having constipation and 53% having FI. The bowel and bladder problems in these diseases are believed to be due to supraspinal or spinal lesions associated with autonomic sympathetic and parasympathetic system impairment.^{19,26,121,144,145}

Spinal Cord Injuries and Disorders

Trauma, masses, infection, hemorrhage, or ischemia to the spinal cord causing tetraplegia or paraplegia primarily affects colonic motility and perineal and anorectal sensation and function in the acute and chronic phases. However, spinal cord–related upper GI tract problems have been more frequently recognized. Depending on the level of the injury, brain and autonomic modulation of the ENS is impaired. The bowel dysfunction is principally one of motility rather than secretion or absorption. It has been demonstrated that the upper digestive tract is more involved in people with tetraplegia than with paraplegia and is characterized by gastroparesis, impaired gastric emptying, and delayed GI transit times. Dysphagia is common in people with cervical cord lesions in the acute phase and often eventually resolves. Gastric and duodenal ulcerations are prevalent in people with myelopathy. Across all levels of injury, there is a loss of voluntary control over bowel movements resulting in constipation and FI. Supraconal lesions present with preserved rectoanal reflexes and spastic sphincters. Infraconal lesions display flaccidity of the sphincters with loss of reflexes.

Peripheral Neuropathy

There are multiple causes of peripheral neuropathy including metabolic, viral, traumatic, toxic, metastatic, and genetic. Sensory and motor deficits are associated with autonomic involvement. Disturbances in GI transit are most often caused by peripheral neuropathies. Motility problems might result in diarrhea (from bacterial overgrowth) or constipation. Persons with diabetes are well known to develop gastroparesis and decreased intestinal motility primarily from autonomic dysfunction. In paraneoplastic syndromes in small cell carcinoma of the lung or pulmonary carcinoid, immunoglobulin G (IgG) antibody against enteric neurons has been found to be responsible for esophageal dysmotility, gastroparesis, and constipation. Injury to the pudendal nerve (from childbirth or protracted straining), S2, S3, or S4 nerve root lesions, or cauda equina underlies FI and abnormal defecation.

Comprehensive Evaluation

History

Recognition of all aspects of impairment and disability that limit a person's ability to maintain continence and volitionally defecate must be assessed. GI history should not only review for cardinal symptoms but should also address the patient's general neuromuscular and GI function (Box 22.7). A detailed review of the patient's bowel program includes an assessment of fluids, diet, activity, medications, and aspects of bowel care. It is imperative to determine defecation frequency, stool consistency (using the Bristol Stool Scale, Fig. 22.6),⁸⁵ stool amount per bowel movement, typical

• BOX 22.7 Components of the Gastrointestinal History

- Cardinal symptoms
- Neuromuscular function
- Gastrointestinal (GI) function
- Total intake of fluids and type of fluids
- Diet: include intake of natural fibers like vegetables and fruits
- Activity
- Comprehensive list of medications
- Oral and rectal GI medications (current and past)
- Frequency of bowel movement (BM)
- Stool consistency (use Bristol Stool Scale)
- Stool amount per BM
- Times of day of BM; schedule of bowel program
- Specifics of bowel program including technique; initiation method, facilitative techniques, time requirements
- Premorbid bowel patterns and presence of disease
- Presence of abdominal discomfort, bloating, distention
- Presence of rectal urgency; ability to prevent stool loss
- Issues with fecal incontinence

Bristol Stool Chart

Type 1		Separate hard lumps, like nuts (hard to pass)
Type 2		Sausage-shaped but lumpy
Type 3		Like a sausage but with cracks on its surface
Type 4		Like a sausage or snake, smooth and soft
Type 5		Soft blobs with clear-cut edges (passed easily)
Type 6		Fluffy pieces with ragged edges, a mushy stool
Type 7		Watery, no solid pieces. Entirely Liquid

• **Fig. 22.6** The Bristol Stool Scale. (From Lewis SJ, Heaton KW. Stool form scale as a useful guide to intestinal transit time. *Scand J Gastroenterol.* 1997;32:920–924.)

time(s) of the day, associated predefecatory activities, oral and rectal bowel medications, and techniques currently and previously used. A review of the technique and outcome of bowel care should include a description of schedule, initiation method (chemical or mechanical stimulation), facilitative techniques, time requirements,

and characteristics of stool results. Careful consideration must be given to drugs that significantly decrease GI motility such as opiates, anticholinergics, tricyclics, antihistamines, calcium-channel blockers, and phenothiazines. The history should include premorbid bowel patterns and GI disease or dysfunction. The presence of GI pain or discomfort, abdominal bloating and distention, warning sensations for defecation, sense of urgency, and ability to prevent stool loss during Valsalva activities such as laughing, sneezing, coughing, or transfers should be noted. Excessively large-caliber hard stool can be ascertained by a history of toilet plugging.⁶⁸ The patient's goals and willingness to alter prior bowel patterns or management need to be established. All aspects of personal performance should be addressed in person-centered rehabilitation, with the overall goal of maximizing independence in bowel management.

For a quick, standardized, valid method of collecting information on bowel function in daily practice, the International SCI Bowel Function Basic Data Set (version 2.0) (Fig. 22.7)⁷⁸ is a useful tool that consists of 16 items related to bowel function, as well as ; the NBD score.⁷⁸ The NBD score assists with determining the severity of bowel dysfunction in people with SCI and is derived from 10 questions based on symptoms. The symptoms are weighted based on their impact on quality of life.⁷⁸ Subsequently, the International SCI Bowel Function Extended Data Set was developed to guide the collection of clinically relevant data in scientific studies on NBD (Fig. 22.8).⁸³

Physical Examination

The physical examination should include the GI system and the associated parts of the musculoskeletal and nervous systems required for adequate management of the bowel program (Box 22.8). The examination should be completed at the initial assessment and then annually for SCI.¹²⁹ The purpose of the examination is to detect functional changes, screen for complications, and identify any new masses or lesions.

The abdomen should be inspected for distention, hernias, and other abnormalities. Careful assessment for the presence or absence of bowel sounds should be done. Percussion and auscultation should precede palpation for masses and tenderness. With the abdomen relaxed, the examiner palpates the abdomen in all quadrants. Palpable hard stool in the colon should be noted and should not be present on the right side of the abdomen (ascending colon). Gastroparesis, intestinal, and colonic pseudo-obstruction present with an abdomen that is tender, distended, and tympanic with hypoactive bowel sounds.⁵⁷ This can be accompanied by autonomic dysreflexia in people with SCI. The patient may show signs of malnutrition and dehydration, including loss of weight, pale skin, dry mucous membranes, poor skin turgor, orthostatic hypotension, and tachycardia.

The rehabilitation evaluation should be interdisciplinary and a process driven by an updated problem list, changes in patient function, alterations in resources, and updated patient-centered goals. It should include an assessment not only for colon and pelvic floor dysfunction, but also for impairments of other organs or systems that could affect rehabilitative strategies to make bowel care independent or prevent unplanned bowel movements.

Physical examination continues with inspection of the anus. A patulous, gaping orifice suggests a history of overdistention and trauma by a previous regimen. A normal anal-buttock contour (see Fig. 22.5) suggests an intact EAS muscle mass, whereas its loss results in a flattened, fanned-out “scaloped” appearing anal region. The patient should perform a Valsalva maneuver while the examiner observes the anus and perineum for excessive descent.

**INTERNATIONAL SPINAL CORD INJURY DATA SETS
BOWEL FUNCTION BASIC DATA SET FORM**

Name: _____ Date performed: _____

Intervention Status: _____

Gastrointestinal or anal sphincter dysfunction unrelated to the spinal cord lesion:

☐ No ☐ Yes, specify _____ ☐ Unknown

Surgical procedures on the gastrointestinal tract:

☐ No ☐ Yes, Appendicectomy, date performed _____
 Cholecystectomy, date performed _____
 Haemorrhoidectomy, date performed _____
 Colostomy, date last performed _____
 Ileostomy, date last performed _____
 Other, specify _____ Date _____

Awareness of the need to defecate (within the last four weeks):

☐ Normal (direct)
☐ Indirect (For example: Abdominal cramping or discomfort - Abdominal muscle spasms - Spasms of lower extremities - Perspiration - Piloerection - Headache - Chills)
☐ None ☐ Unknown

Defecation method and bowel care procedures (within the last four weeks):

	Main	Supplementary
Normal defecation	☐	☐
Straining/bearing down to empty	☐	☐
Digital ano-rectal stimulation	☐	☐
Suppositories	☐	☐
Digital evacuation	☐	☐
Mini enema (Clysmas ≤150 mL)	☐	☐
Enema (>150 mL)	☐	☐
Colostomy	☐	☐
Sacral anterior root stimulation	☐	☐
Other method, specify _____	☐	☐
☐ Unknown		

Average time required for defecation (within the last four weeks): *

☐ 0–5 min ☐ 6–10 min ☐ 11–20 min ☐ 21–30 min ☐ 31–45 min³ ☐ 46–60 min³
☐ 61–90⁷ ☐ 91–120⁷ ☐ 121–180⁷ ☐ More than 180 min⁷ ☐ Unknown

Frequency of defecation (within the last four weeks):*

☐ Daily ₍₀₎
☐ Every other day ₍₁₎
☐ Twice every week ₍₁₎ ☐ Once every week ₍₆₎
☐ Less than once every week, but at least once within the last four weeks ₍₆₎
☐ No defecation within the last four weeks ₍₆₎ ☐ Not applicable ☐ Unknown

Uneasiness, headache, or perspiration during defecation (within the last four weeks):*

☐ No ₍₀₎ ☐ Yes ₍₂₎ ☐ Unknown

Digital stimulation or evacuation of the anorectum (within the last four weeks): *

☐ Daily ₍₆₎
☐ Once or more per week ₍₆₎
☐ Less than once per week ₍₀₎
☐ Never ₍₀₎ ☐ Not applicable ☐ Unknown

• **Fig. 22.7** International Spinal Cord Injury Bowel Function Basic Data Set (Version 2.0). Bowel function basic data sets. *NBD*, Neurogenic bowel dysfunction. (Modified from Krogh K, Emmanuel A, Perrouin-Verbe B, et al. International spinal cord injury bowel function basic data set [Version 2.0]. *Spinal Cord*. 2017;55:692–698.)

Frequency of fecal incontinence (within the last four weeks):*

☐ Daily ⁽¹³⁾
☐ 1-6 times per week ⁽⁷⁾
☐ 1-4 times every month ⁽⁶⁾
☐ Less than once per month ⁽⁰⁾
☐ Never ⁽⁰⁾

Flatus incontinence (within the last four weeks):*

☐ No ⁽⁰⁾ ☐ Yes ⁽²⁾ ☐ Unknown

Need to wear diaper, pad, or plug (within the last four weeks):

☐ No ☐ Yes ☐ Unknown

Oral laxatives (within the last four weeks):*

☐ No ⁽⁰⁾ ☐ Yes, osmotic laxatives (drops)
☐ Yes, osmotic or bulking laxatives (tablets or granulates)
☐ Yes, irritant laxatives (drops)
☐ Yes, irritant laxatives (tablets)
☐ Yes, prokinetics
☐ Yes, other, specify: _____ ⁽²⁾
(e.g., osmotic: lactulose, psyllium, magnesium, sorbitol; irritants: Bisacodyl (Dulcolax), sodium picosulfate (Prepopik))

☐ Unknown

Constipating agents / drugs against fecal incontinence (within the last four weeks):*

☐ No ⁽⁰⁾ ☐ Yes ⁽⁴⁾ ☐ Unknown

Perianal problems (within the last year):*

☐ No ⁽⁰⁾ ☐ Yes ⁽³⁾
☐ Hemorrhoids ☐ Perianal sores ☐ Fissures
☐ Other, specify _____ ☐ Unknown

Abdominal pain or discomfort (within the last four weeks):

☐ Daily ☐ 1-6 times per week ☐ Once per week or less
☐ Never ☐ Unknown

General satisfaction with your bowel management:

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10
 (Total dissatisfaction = 0 / Perfect satisfaction = 10)

Impact of your bowel management on your daily life:

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10
 (Total disruption = 0 / No impact whatsoever = 10)

General Comments / Other Relevant Information:

Total NBD Score:

☐ 0-6 Very minor
☐ 7-9 Minor
☐ 10-13 Moderate
☐ 14 or more Severe neurogenic bowel dysfunction

* Items used for computation of the NBD score.

*Adapted (Nov. 2013; edited July 2018) by C.H. Hubscher from:
 K. Krogh et al. (2017) Spinal Cord 55: 692–698 (version 2)
 K. Krogh et al. (2009) Spinal Cord 47: 230–234.
 K. Krogh et al. (2006) Spinal Cord 44: 625–631*

• Fig. 22.7, cont'd

Date: _____

The Neurogenic Bowel Dysfunction Score

1. How often do you defecate? ☐ Daily (score 0) ☐ 2-6 times per week (score 1) ☐ Less than once per week (score 6)	Score
2. How much time do you spend on each defecation? ☐ Less than 30 min. (score 0) ☐ 31-60 min. (score 3) ☐ More than an hour (score 7)	
3. Do you experience uneasiness, sweating, or headaches during or after defecation? ☐ Yes (score 2) ☐ No (score 0)	
4. Do you take medication (tablets) to treat constipation? ☐ Yes (score 2) ☐ No (score 0)	
5. Do you take medication (drops or liquid) to treat constipation? ☐ Yes (score 2) ☐ No (score 0)	
6. How often do you use digital evacuation? ☐ Less than once per week (score 0) ☐ Once or more per week (score 6)	
7. How often do you have involuntary defecation? ☐ Daily (score 13) ☐ 1-6 times a week (score 7) ☐ 3-4 times a month (score 6) ☐ A few times a year or less (score 0)	
8. Do you take medication to treat fecal incontinence? ☐ Yes (score 4) ☐ No (score 0)	
9. Do you experience uncontrollable flatus? ☐ Yes (score 2) ☐ No (score 0)	
10. Do you have peri-anal skin problems? ☐ Yes (score 3) ☐ No (score 0)	
Total score (between 0 and 47)	

General satisfaction

Please mark the scale with a cross (x) to represent your general satisfaction with your bowel management.
(Total dissatisfaction = 0 / Perfect satisfaction = 10)

☐ 0
 ☐ 1
 ☐ 2
 ☐ 3
 ☐ 4
 ☐ 5
 ☐ 6
 ☐ 7
 ☐ 8
 ☐ 9
 ☐ 10

Severity of bowel dysfunction

Score 0-6: Very minor
 Score 7-9: Minor
 Score 10-13: Moderate
 Score 14+: Severe

• **Fig. 22.8** The neurogenic bowel dysfunction score. (From Krogh K, Christensen P, Sabroe S, et al. Neurogenic bowel dysfunction score. *Spinal Cord*. 2006;44:625–631.)

Perianal cutaneous sharp stimulation normally results in an externally visible anal sphincter reflexive contraction. This is the anocutaneous reflex, mediated by the inferior hemorrhoidal branch of the pudendal nerve (S2 to S5). Sensation to light touch and pinprick is tested at the same time. If deep anal pressure is perceived in SCI, this denotes incompleteness of the SCI on the International Standards for Neurological Classification of Spinal Cord Injury examination.⁶⁹ The tone and voluntary squeeze strength of the EAS and tone of the IAS should be assessed. SCI is noted to be American Spinal Injury Association Impairment Scale C (AIS C)

• BOX 22.8 Components of the Physical Examination

Abdominal Evaluation

- Observation—signs of malnutrition, dehydration, anemia
- Inspection—distention, hernias
- Auscultation—bowel sounds
- Percussion—observe for tympanitic abdomen
- Palpation—tenderness, hard stool

Perineal/Anorectal Evaluation

- Inspection—for gaping orifice or puckered anal sphincter
 - Contour of buttocks
 - Hemorrhoids, fissures, bleeding
 - Visual observation of anal squeeze and perineal relaxation and descent with Valsalva maneuver
- Evaluate for anocutaneous reflex (anal wink); bulbocavernosus reflex
- Examination of sensation to light touch and pinprick; sensation to deep anal pressure
- Digital rectal examination
 - Sphincter tone at rest
 - Ability to do a voluntary anal contraction and rectal squeeze
 - Ability to resist defecation
 - Ability to relax sphincter and pelvic floor and push finger with simulated defecation
 - Evaluate for paradoxical contraction of the sphincter with pelvic floor relaxation during simulated defecation

if this is present,⁶⁹ implying better prognosis for motor function. Integrity of the pelvic floor muscles can be examined by the ability to contract and relax. The length of the anus, where pressure is sensed, is normally 2.5 to 4.5 cm. The point where pressure decreases marks the anorectal junction. Along the posterior wall, 1.5 to 2.5 cm from the anal verge, the puborectalis muscular sling can be palpated as a ridge that will push the finger forward toward the pubis as the subject resists defecation. No palpable ridge or push suggests puborectalis atrophy or dysfunction. A shortened length of anal pressure zone suggests EAS muscle atrophy. With the examiner's finger in place, the bulbocavernosus reflex can be elicited by rapidly tapping or squeezing the clitoris or glans penis. The response can be delayed up to a few seconds in pathologic conditions. Insertion of the finger in the anal canal occasionally triggers IAS relaxation but more often triggers a tightening EAS contraction that is efferently equivalent to the bulbocavernosus reflex. The bulbocavernosus reflex (and the anocutaneous reflex) is weak or absent when in spinal shock or in infraconal lesions. Ask the patient to volitionally squeeze the anus before removing the finger ("resist defecation") to check for volitional EAS and puborectalis tone and control. In addition, ask the patient to bear down and relax alternately to evaluate contraction, relaxation, and coordination of the pelvic muscles.

Diagnostic Testing

Additional objective laboratory testing can be helpful when the cause of FI or DWE is obscure, the history appears doubtful, conservative interventions fail, or surgical interventions are contemplated. Table 22.2 lists some of the many tests available.

Radiographic testing complements the physical examination (Box 22.9). A flat plate radiograph with or without an upright of the abdomen can be helpful to evaluate for the degree of fecal loading, rule out impaction, megacolon, obstruction, and a perforated viscus.^{12,15} If further indicated, a computed tomography (CT) scan of the abdomen is a reliable, specific, and sensitive method in assessing detailed structural issues like small or large intestinal obstructions and other lesions.^{12,15}

TABLE 22.2 Clinical Assessments for Neurogenic Bowel

Test	Description	Additional Comments and Recommendations for Use in SCI/D
Bowel Diary	A patient-generated time-stamped log of bowel action. Records stool type (Bristol Stool Scale), amount, incontinence episodes, urgency, straining and all current medications (not just those targeting bowel function). Complications such as nausea or hemorrhoids can be captured in comments.	Use to evaluate bowel management routine, characterize daily habits/lifestyle choices and document gastrointestinal symptoms. Recommendations: • Include documentation of time taken for bowel care episode from start to finish. • Uncover patient perception of ideal/reasonable timing and frequency of bowel evacuation for comparison to current experience. • Also include assistance such as physical interventions (i.e. digital stimulation, disimpaction, enemas, etc.), position (in bed or over toilet), caregiver support, and use of suppositories or oral medications in terms of type, amount and timing related to bowel movement. • International Spinal Cord Bowel Function Basic Data Set (version 2.1) and the Neurogenic Bowel Dysfunction Score (NBDS) can be used.

TABLE 22.2 Clinical Assessments for Neurogenic Bowel—cont'd

Test	Description	Additional Comments and Recommendations for Use in SCI/D
Physical Examination of Bowel Dysfunction	Conducted by a clinician and includes abdominal examination, rectal ampulla exam (for hard stools or tumors), tests of sacral reflexes (anocutaneous and/or bulbocavernosus), anal sphincter tone, sensation and voluntary contraction.	Use to identify bowel dysfunction related conditions. Recommendations: • Important to determine the level of injury and verify upper vs. lower motor neuron deficits. • Verify SCI/D pelvic floor muscle function during defecation for role in defecation disorder. • Auscultation of the abdomen followed by palpation and percussion should be included. • In patients with an ostomy, include examination of stoma and check for direct or indirect hernia.
Stool Sampling for Analysis	A series of tests to help diagnose certain conditions affecting the digestive tract. Stool sample often collected by the patient at home.	Use when screening for cancer, infection, or suspicion of parasites. Recommendations: • Tests for toxins or leukocytes may be included if concern about infection. • Consider evaluation of guaiac or fecal occult blood (starting at age 50 or sooner if problems arise).
Abdominal X-ray	Standard testing to evaluate extent and location of fecal loading.	Use to determine of megacolon, degree of constipation, and site of impaction as well as the need for a more aggressive bowel management strategy. Recommendations: • Helpful to determine if oral versus rectal route of treatment is needed based upon site of impaction.
Computed Tomography (CT) Scan	Standard radiologic imaging helpful in delineating gastric, small intestinal, colonic or pelvic structural or anatomic abnormalities.	Use to exclude lesions that may cause gastrointestinal symptoms and constipation.
Endoscopy (Rectosigmoidoscopy, Anoscopy, and Colonoscopy)	Standard test if cancer or structural abnormalities are suspected.	Use to diagnose or rule out colon abnormalities and carcinoma.
Gastric Emptying Study	Test to determine the time it takes for food to move through a person's stomach.	Useful in evaluating for gastroparesis in persons with SCI/D experiencing significant upper abdominal symptoms such as abdominal pain/discomfort, bloating, and distention and treating accordingly.
Total and Segmental Colonic Transit Time (CTTs)	Test that uses radio-opaque markers to investigate small and large bowel motility through objective time course measurement.	Use in determining worsening colonic function and if there is a need for surgical intervention. Recommendations: • Although easy to perform, CTT should only be used to support clinical evaluation as large variations in transit time normally exist which reduces its diagnostic value.
Wireless Motility Capsule for Bowel Transit	Test that involves ingesting a capsule that simultaneously measures the gastrointestinal and colonic pH, temperature and intraluminal pressure to provide information on gastric emptying, small intestine transit and total colorectal transit.	Use to gain comprehensive information on gastrointestinal motility and how motility of each gut segment is correlated to the other (gastric vs small intestinal vs colon). Recommendations: • Consider use only in SCI/D patients with intact swallowing capability. • Not universally available.
Anorectal Manometry ± Balloon Expulsion Test (BET)	Test using a balloon in the rectum and a pressure sensor at the sphincter to measure contractility in the anus and rectum.	Use to determine internal anal pressure and the presence or absence of the rectosphincteric reflex. Recommendations: • Can be very helpful in delineating defecation disorders related to pelvic floor dysfunction in incomplete SCI/D. • Can provide information of anorectal physiology including quantitative measures of rectal volume sensations and anorectal pressure during rest and squeeze that is useful for persons with incomplete SCI.

Continued

TABLE 22.2 Clinical Assessments for Neurogenic Bowel—cont'd

Test	Description	Additional Comments and Recommendations for Use in SCI/D
Pudendal Nerve Latency Testing and Pelvic Floor Electromyography	Electrophysiological examination used to directly investigate the integrity of the somatic innervation of the pelvic floor muscles and urinary and anal sphincters.	Use to evaluate defecation disorder and pelvic floor dysfunction. Recommendations: • Helpful in evaluation of peripheral nerve injuries in incomplete SCI/D and peripheral neuropathy (especially pudendal neuropathy). • Important for consideration of the level of injury and to determine upper vs. lower motor neuron deficits.
Defecography	Medical radiological imaging test in which the mechanics of a patient's defecation are visualized in real time using a fluoroscope.	Use when anatomic causes of outlet obstruction like rectal prolapse, rectocele or enterocele are suspected. Recommendations: • Helpful for assessment of progressive worsening or change in bowel patterns in incomplete SCI.

Modified from Tate DG, Wheeler T, Lane GI, et al. Recommendations for evaluation of neurogenic bladder and bowel dysfunction after spinal cord injury and/or disease. *J Spinal Cord Med.* 2020;43(2): 141–164. Table 3.

• BOX 22.9 Basic Diagnostic Testing

- Basic laboratory tests
- Abdominal radiography
- Computed tomography abdomen
- Colon transit testing/wireless motility capsule
- Anal rectal manometry, balloon expulsion test $\pm$ electromyography/pudendal nerve testing

Colon transit testing requires swallowing radiopaque markers that are followed by abdominal radiographs as they pass through the various parts of the colon. Colon transit time is measured by counting the passage of markers through the right, transverse, and rectosigmoid segments of the colon.^{12,141} Studies completed on people with NBD due to SCI have shown that transit time is primarily delayed in the left colon and rectum, but may be pancolonic.^{35,77,137} The Wireless Motility Capsule has been shown to be an essential method for measuring GI transit time for NBD with the crucial advantage of measuring segmental gastric, small intestinal, and colonic transit time all with one test compared with conventional testing done with separate tests, on different days.¹⁴⁶

For people who have neurologic issues with associated constipation/FI and who have present but impaired sensation and/or motor function, the anal rectal manometry test is helpful in characterizing a defecatory disorder (Fig. 22.9).^{12,116,141} Supraconal NBD may present as dyssynergia (paradoxical contraction of the rectal sphincter) and/or failure to relax pelvic floor muscles during simulated defecation, with less than 20% decrease in anal canal pressures.^{77,139} It may be associated with inadequate defecatory propulsion due to inadequate increase in rectal or intraabdominal pressure during simulated defecation.¹³⁹ Intraconal injuries will present alternatively with decreased rectal reflexes and decreased resting and volitional sphincter pressures.^{77,139}

Anorectal manometry is typically accompanied by balloon expulsion testing (BET), which involves evacuation of a lubricated balloon-tipped catheter. BET is known to have high specificity for dyssynergia. Balloon expulsion time is abnormal if greater than 1 minute.^{12,116,141} BET has been used to further evaluate defecatory disorders in NBD.^{77,93,139}

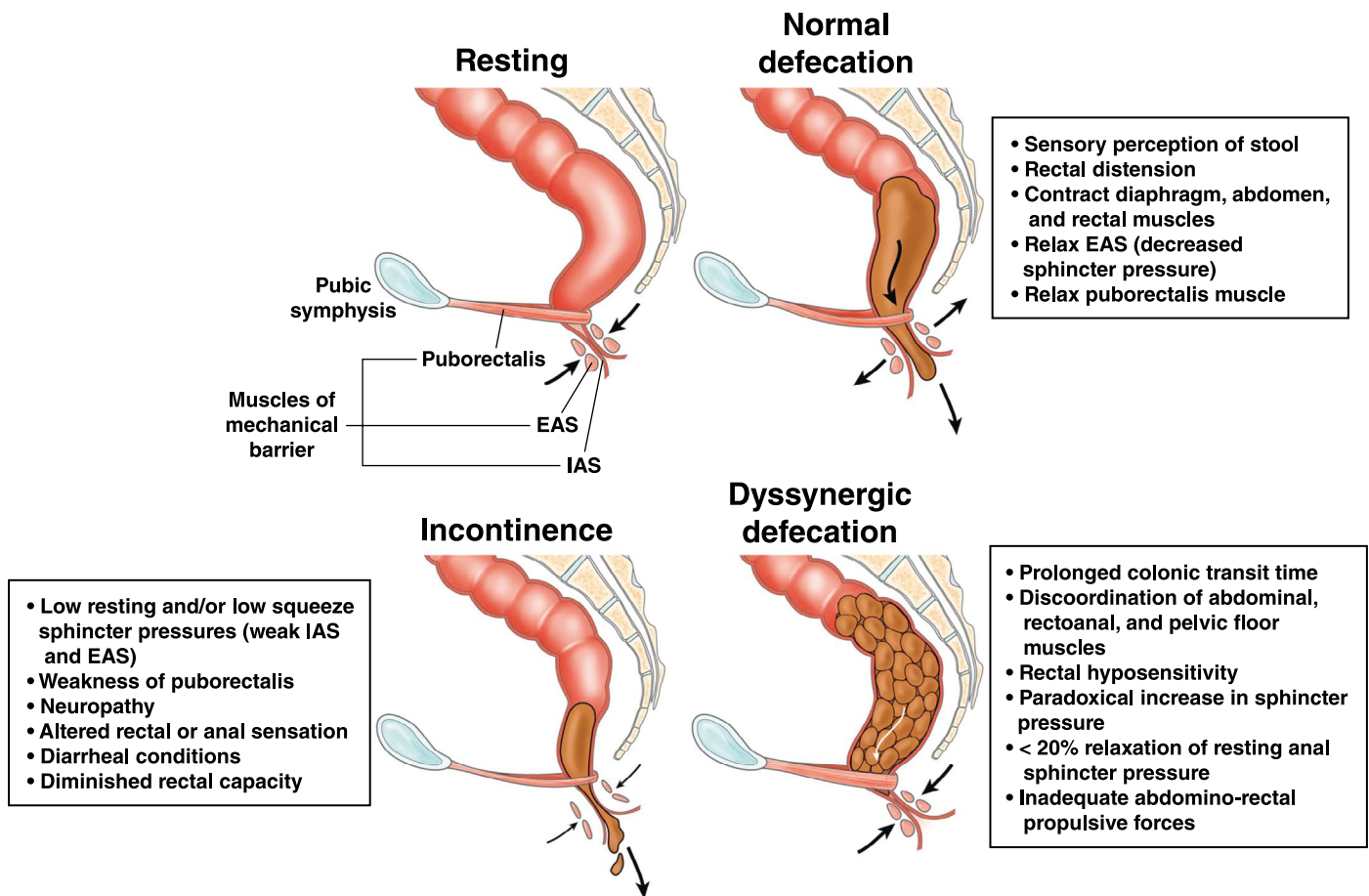
EMG activity can be recorded with electrodes applied to bilateral areas of the rectum to augment anorectal and pelvic floor testing.^{12,116,141} It is another measure used to detect dyssynergia or characterize defecation dysfunction in NBD. It is often performed with pudendal nerve latency testing, which assists with identifying peripheral pudendal nerve injury.^{46,77,139,143}

Defecography, using fluoroscopy or magnetic resonance imaging, is a dynamic assessment of the anorectum and pelvic floor muscles before, during, and after attempted defecation of barium paste applied in the rectum. The puborectalis muscle can be assessed for contraction (dyssynergia) and relaxation (normal) during simulated defecation by indirectly measuring the angle between the axis of the rectum and anal canal. Defecography is most useful for evaluation of structural causes of outlet dysfunction like rectal prolapse, rectocele, and enterocele.^{12,116,141}

Management

Management of Nausea, Vomiting, Bloating, and Early Satiety

Nausea, vomiting, bloating, and early satiety may indicate gastroparesis; a small intestinal or colonic pseudo-obstruction in people with neurologic disease.⁵⁷ Acute or chronic episodes of subocclusion, characterized by derangement of gut propulsive motility resembling mechanical obstruction in the absence of any obstructive process, can occur. There may be findings of distended bowel loops and air-fluid levels requiring GI decompression with nasogastric or rectal tubes. The patient may be placed on bowel rest and, to reduce gut irritation, consume nothing by mouth. Nutritional, fluid, and electrolyte support must be provided with intravenous infusion. Drugs such as opiates, anticholinergics, tricyclics, antihistamines, calcium-channel blockers, and phenothiazines (all of which profoundly diminish motility) must be minimized or discontinued. Once decompression is achieved, feeding is slowly advanced to a general diet. Prokinetic agents such as metoclopramide and domperidone (dopamine antagonists) and erythromycin (macrolide antibiotic, motilin receptor agonist) have been found to be beneficial in facilitating gastric emptying and



• **Fig. 22.9** This series of schematic diagrams reveals the normal anatomy and physiology of the pelvic floor in the sagittal plane at rest, during defecation, and the key pathophysiologic changes in subjects with fecal incontinence and dyssynergic defecation. The resting profile shows the arrangement of the three anal muscles including the puborectalis muscle and the normal changes that occur during defecation: The anal sphincters and the puborectalis relax, and the abdominal and rectal pushing forces generate a push force to empty the rectum and stool. In a patient with fecal incontinence, the anal and puborectalis muscle may be weak, causing stool leakage. In subjects with dyssynergia, the push effort or anal relaxation may be impaired or the anal sphincters and puborectalis may not relax during attempted defecation, leading to incoordination (dyssynergia) and stool retention. *EAS*, External anal sphincter; *IAS*, internal anal sphincter. (From Rao SS. Advances in diagnostic assessment of fecal incontinence and dyssynergic defecation. *Clin Gastroenterol Hepatol*. 2010;8:910–919.)

intestinal motility and improving symptoms. A recent review of studies with high-quality randomized clinical controlled trials showed good results for prucalopride (now available in the United States), neostigmine (administered both with and without glycopyrrolate), and fampridine.⁷⁶ For chronic subocclusive states, enteral nutrition is preferred in patients with satisfactory digestive function. Meals should be small, frequent, and liquid; contain polypeptide and hydrolyzed protein; be low in fat, fiber, and lactose; and contain multivitamins (iron, folate, calcium, vitamin D, K, B₁₂). If oral intake is not tolerated, enterostomies can be helpful to optimize nutrition. Total parenteral nutrition to satisfy nutritional requirements should be reserved for very severe cases. Potential complications such as pancreatitis, glomerulonephritis, liver abnormalities, and septicemia or thrombosis from chronic catheter use must be considered.

During brief periods, small intestinal bacterial overgrowth (SIBO) syndrome may present with nausea, vomiting, bloating,

and early satiety. It can also present with diarrhea in chronic inflammatory conditions. This occurs when there is an overwhelming increase in bacteria or other pathogens in the upper GI tract (stomach, duodenum, jejunum, and proximal ileum), where there is normally low amounts of bacteria.¹¹⁷ There are many causes of bacterial overgrowth syndrome, but with neurologic disease, it is most likely due to upper GI tract dysmotility leading to decreased clearance of unwanted bacteria and undigested substances.¹⁶ To diagnose this condition, the glucose hydrogen breath test is the simplest to perform. Levels of exhaled hydrogen to 20 ppm indicate a positive test. This test is 82% sensitive and 62.5% specific.⁴⁴ Treatment with rifaximin (550 mg two to three times per day) is recommended.⁴² A combination of ciprofloxacin and metronidazole has also been shown to be effective.^{7,20}

Diarrhea in neurologic disease is commonly due to severe constipation. When solid stools are not passed regularly, they become very hard and are difficult to pass, causing a blockage of waterier

stool behind them. Eventually, there is overflow of the watery stool around the hard stool, causing diarrhea. Therefore it is necessary to determine if diarrhea is actually a result of constipation through history, physical examination, and abdominal x-ray to evaluate for fecal impaction or fecalith.

With repeated use of antibiotics for treatment of urinary tract infections, pneumonia, and other infections, diarrhea is frequently due to *Clostridium difficile* colitis. It can be associated with crampy abdominal pain, anorexia, malaise, lower abdominal tenderness, and, in more severe cases, fever and dehydration. A stool test confirms the presence of the toxin or the cytotoxin.^{5,41} Treatment with metronidazole is used to treat mild to moderate colitis. Oral vancomycin is used to treat severe or complicated disease producing faster symptom resolution and fewer treatment failures. In fulminant cases, combined therapy with intravenous metronidazole and oral (or per rectum) vancomycin is recommended.⁶¹ Fecal transplantation is currently a viable and effective option in the treatment of recurrent colitis.¹¹⁴

More recently, there has been increasing use of probiotics, which are live nonpathogenic bacteria (*Lactobacillus*, *Bifidobacterium*) or yeast (*Saccharomyces boulardii*), to treat illnesses that result in diarrhea. Probiotics are thought to block production of microbial toxins, inhibit pathogen adhesion, modulate the immune system, and improve the gut barrier function. Probiotics were shown in a Cochrane Review to reduce duration of illness with diarrhea lasting less than 4 days and decrease in stool frequency.⁹⁸

Management of Defecation Dysfunction: Constipation and Fecal Incontinence

Management of NBD involves a pyramidal stepwise method, which commences with a personalized bowel program and conservative treatment composed of minimizing constipating medications, good intake of fluids and fiber, oral laxatives, and rectal medications; and mechanical strategies for bowel evacuation (Fig. 22.10).^{36,54,76}

A *bowel program* is a comprehensive individualized patient-centered treatment plan focused on preventing incontinence,

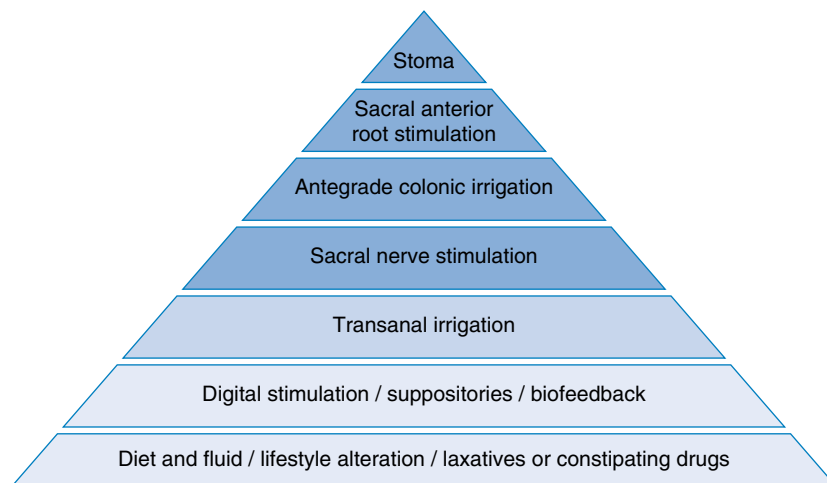
achieving effective and efficient colonic evacuation, and preventing the complications of NBD. The subcomponents of a bowel program address diet, fluids, exercise, medications, and scheduled bowel care. *Bowel care* is the individually developed and prescribed procedure for defecation that is carried out by the patient or attendant.^{67,129}

NBD results in problems with fecal storage and elimination. Inability to volitionally inhibit spontaneous defecations leads to incontinence while the inability to adequately empty leads to constipation and impactions. Paradoxically, social continence is achieved by impaction and can result in diarrhea and incontinence. Providing adequate timely emptying must be combined with the inhibition of spontaneous defecations except at desired times.¹³⁰ The individual life roles of the patient that are achieved after completing the acute rehabilitation process determine the bowel program and bowel care schedule. The demands of life activities, the duration of bowel care, attendant care options, and the needs of other members of the household should all be considered when designing the bowel program and the process of bowel care.^{70,129}

Goals of the Bowel Program

It is essential to outline the goals of the bowel program as a person-centered plan for each patient (Box 22.10). Each person responds differently to various techniques and medications used to treat constipation and FI. Regular bowel emptying is recommended as the primary means for enhancing both elimination and decreasing incontinence between stooling. The goals of the bowel program should include^{12,117}:

1. Regular, planned passage of stool on a daily or every other day basis (at least 3 times per week)
2. Adequate stool outputs per bowel movement (moderate-large stools)
3. Bowel evacuation at a consistent time of day (AM or PM)
4. Complete emptying of the rectal vault with every bowel care session
5. Stools that are soft, formed, and bulky



• **Fig. 22.10** A proposed stepped approach to treatment of bowel dysfunction. The lower two layers represent “conservative” treatment, the middle layer represents “minimally invasive” treatment, and the four upper layers represent “more invasive” treatment options. (Redrawn from Emmanuel AV, Krogh K, Bazzocchi G, et al. Consensus review of best practice of transanal irrigation in adults. *Spinal Cord*. 2013;51:732–738.)

• BOX 22.10 Goals of the Bowel Program

- Regular, planned passage of stool on a daily or every other day basis (at least 3x per week)
- Adequate stool outputs per bowel movement (moderate-large stools)
- Bowel evacuation at a consistent time of day (AM or PM)
- Complete emptying of the rectal vault with every bowel care session
- Stools that are soft, formed, and bulky
- Completing the bowel care within half an hour (at most within 1 hour)
- No episodes of fecal incontinence or unplanned bowel movements

6. Completion of bowel care within half an hour (at most within 1 hour)
7. No episodes of fecal incontinence or unplanned bowel movements

Fully appreciating an individual's premorbid "normal" bowel function is important in the planning and goal setting for a new neurogenic bowel program. A wide range of "normal" bowel patterns exist. Ninety-five percent have a frequency of between three times per day and three times per week. Stool consistencies vary from liquid to pudding, pasty, semisolid, soft formed, medium formed, and hard formed.^{67,70}

The choice of the frequency and specific timing of bowel care to induce adequate colonic emptying can be based on previous elimination patterns. However, it is best to plan for every day or every other day bowel movements to avoid episodes of FI or impaction. Waiting more than 2 days to have a bowel movement increases episodes of constipation and incomplete emptying of the rectum. Establishing a consistent schedule for bowel care facilitates predictability and planning. Ensuring complete emptying of the rectum decreases incontinence or stooling during the course of the day. Incontinence is reduced by less stool accumulation because stool is not presented to the rectum between desired defecation times. Adequate emptying is accomplished by making stools easier to propel by softening, adding bowel stimulant medication if needed, and triggering the defecatory reflex at consistent desired times to promote habituation. If conal and local rectal reflex methods do not deliver stool, manual evacuation or various irrigating techniques are effective.^{67,70,130}

Bowel programs and techniques for bowel care training should begin during inpatient rehabilitation and then be monitored, maintained, or modified at outpatient visits. Some patients need attendants who are well trained to help with bowel care.¹²⁹ The burden of care for persons with neurogenic bowel is much higher if continence is not achieved or if bowel care evacuation times are excessive. It is crucial that the patient takes a decisive leadership role in designing a bowel program that includes a convenient bowel care schedule. Educating patients about their altered neurogenic bowel physiology empowers them with options and techniques to construct a bowel care regimen compatible with their daily routine. Making them independent with their bowel care is an important aspect of the overall rehabilitation process.^{67,70,129}

Dietary Considerations

Food and fluid choices are important when colonic transit time is prolonged, such as in SCI where the transit time is often 96 hours versus the 30 hours typically found in normal subjects. The main goal of dietary choices is to achieve soft, well-formed, bulky stools. Excessive fluid resorption can result in stool hardening and

subsequent constipation. Gases and liquids are propelled by the colon 30 to 100 times faster than solids. Stools that have lost their plasticity might not be kneaded and folded properly by the haustra, impeding the transit time.

Food fiber has been used to maintain a more fluid content. Fiber increases stool bulk and plasticity, especially in the more physically coarse forms, which also tend to decrease colonic pressures.¹⁵³ Control of excessive stool hardness requires higher-fiber foods in preference to lower-residual foods. However, it is important to remember that sufficient intake of fluids (preferably water) is imperative when taking a high-fiber diet. The Institute of Medicine recommends daily fluid intake of 3 L for males and 2.2 L for females.¹¹¹ Otherwise, a high-fiber diet with insufficient intake of fluids can promote constipation. It should be remembered that highly caffeinated fluids such as coffee, tea, and energy drinks can lead to dehydration from diuresis.

A diet that contains at least 38 g for males and 25 g for females of fiber daily is recommended (15 g initially and gradually increase as tolerated).^{62,130} Aside from fiber from vegetables, fruits, and grains, processed fiber products can be used such as psyllium (Metamucil, Fiberall), calcium polycarbophil (Fibercon), and methylcellulose (Citrucel). One study showed that taking psyllium for 8 weeks increased the number of stools per week and stool weight in patients with Parkinson disease without affecting colonic transit time. A case series study proposes that increase in fiber in individuals with SCI may not improve, but impair bowel function compared with people with normal bowels.^{17,76} The effects of fiber intake on stool consistency, frequency, and efficacy of evacuation should be evaluated in each individual patient, and fiber intake should be titrated accordingly. High pressures involved in moving solid feces probably contribute to the 90% incidence of hemorrhoids in Americans and to premature diverticula formation and hemorrhoidal complications in more than 70% of SCI patients.¹²⁹ Constant straining at stool can contribute to peripheral neuropathic deficits in the anal sphincter musculature.¹²⁹ Softer stools, from a higher-fiber diet, might help to reduce these complications.

Treatment Approaches and Rationale

To select the best approach, the supraconal NBD must be delineated from the infraconal NBD. Brain disease and spinal cord disorders above the conus present with reflexic NBD. Peripheral neuropathy and spinal cord disorders below the conus present with hyporeflexic NBD. Colonic transit time and fecal elimination are enhanced by softer stool. However, if the stool becomes too liquid, the protective angle provided by the puborectalis becomes less effective, and greater EAS pressures are required to maintain continence. Neurogenic bowel resting anal pressures are usually normal to slightly decreased, but patients are unable to develop the protective increases in EAS tone needed to control more liquid stool.¹⁰⁵ Some degree of stool firmness must be tolerated to prevent incontinence. To avoid incontinence upon straining, more firmness (medium formed) is required for the weaker anal sphincter mechanism of infraconal NBD than for supraconal NBD (semi formed to soft formed).

Fiber more consistently softens stool but also adds bulk. Bulkier stools can help to prevent liquefaction of stool and maintain continence for supraconal NBD. Fiber maintains consistency and ease of propulsion and prevents left colonic compaction with constipation. Adequate fluid intake is required for fiber to be helpful; otherwise, fiber can contribute to constipation.⁶²

• BOX 22.11 Protocol for Progressive Steps in Bowel Habituation Program

1. Perform bowel cleanout if stool is present in the rectal vault or palpable proximal to the descending colon by multiple enemas or oral cathartic.
2. Titrate to soft stool consistency with diet and bulking agents (fiber) and stool softeners (docusate).
3. Trigger defecation with a bisacodyl suppository or by digital stimulation 20 to 30 minutes after a meal; 10 minutes later, have the patient attempt defecation on the toilet, limited to <40 minutes and relieving skin pressure every 10 minutes.
4. If defecation is not initiated, a trial of a bisacodyl suppository per rectum is initiated.
5. Digital stimulation. Start 15 minutes after suppository placement and repeat every 5 minutes.
6. Timed oral medications. Administer docusate (Colace) or senna–docusate sodium (Peri-Colace), senna (Senokot), or bisacodyl (Dulcolax) tablets timed approximately 6 to 8 hours prior (adjust accordingly) so that bowel movement would otherwise result 30 minutes to 1 hour after anticipated triggered bowel timing.
7. If defecation occurs in <10 minutes after suppository insertion, transition to digital stimulation technique only. Once the patient is well habituated, straining alone may rarely trigger defecations at a desired time.

For a bowel program to be successful, a combination of oral and rectal medications and bowel evacuation techniques might be required to facilitate more consistent, regular bowel emptying. Stool must effectively pass through the entire length of the colon (approximately 5 feet) for a successful bowel movement to occur. Careful consideration of how medications affect stool consistency is critical to avoid issues with FI.

Rapid advancement of stool to the rectum, triggering defecation, may be due more to GMC and mass movements than with the slow accumulation of sufficient rectal stool to trigger reflex defecation. The GMC might be what is actually habituated.¹³¹ Choosing long intervals between elimination allows more fluid reabsorption, resulting in harder stools, which can worsen DWE. Because 95% of unimpaired persons defecate three or more times per week, choosing a frequency of at least every other day would seem more physiologic and less likely to contribute to constipation. One study of well-managed SCI patients found frequencies of bowel care to be chosen as daily by 24%, every other day by 46%, and more often than three times a week by 85%. The desire to avoid the unpleasant task of stool elimination leads some to elect longer time intervals between bowel care sessions. This carries the risk for impaction or sphincter damage caused by rectal distention by larger-caliber, harder stools.

One approach to initiating neurogenic bowel training is outlined in [Box 22.11](#). Each step is added only after a 7-day consistent trial of the previous step has been ineffective. In this stepwise approach, obtaining elimination at the desired time is emphasized as the first step and usually precedes development of complete continence by several weeks. This regimen is designed to enhance responsiveness and emptying at the habituated time with less stool presentation between bowel care sessions.

Management of Reflexic Supraconal Neurogenic Bowel Dysfunction

In supraconal NBD, the defecatory reflex is intact, so that triggering of defecation can be accomplished by digital stimulation,

rectal stimulant medications, enemas, or electrical stimulation. All of these cause reflex relaxations of the IAS and, if strong enough, can reflexively relax the EAS as well. This activates preserved anorectal colonic reflexes that increase motility of the left colon and help to eliminate any stool that is present.^{74,76} The GMC and mass movement associated with the “call to stool” for many intact persons often occurs at consistent times, which can be trainable.

Digital rectal stimulation is done by inserting a gloved, lubricated finger into the rectum and stimulating the rectal wall with gentle circular movements. This should be done for 20 seconds and repeated every 5 to 10 minutes until the bowel care is completed and the rectal vault is emptied.

Management of Hyporeflexic Infraconal Neurogenic Bowel Dysfunction

In infraconal NBD, the bowel’s reflex branches to and from the conus are interrupted and the most effective way of completely emptying the rectum is manual disimpaction with Valsalva and gravity. The process can be enhanced with lubrications, enemas, and mineral oil or with the use of cleansing enemas (water, soap-suds, mineral oil, milk of molasses) that are done daily or twice a day. Sphincter and pelvic floor muscle tone is lost, increasing the likelihood of FI. Consequently, it is imperative to keep the stools well-formed and bulky (i.e., BSS4) and to empty the rectal vault more regularly. Use of oral medications as outlined for the supraconal NBD might likewise be warranted if stool is not efficiently reaching the rectum in a timely manner.

Intrinsic loss of the ENS in any segment, or surgical reanastomosis, can result in loss of the rectoanal inhibitory reflex, causing DWE. Oral laxative abuse can cause dysfunction of the ENS. If bowel training is not accomplishing defecation at the desired times or if repeated involuntary incontinence occurs, further diagnostic evaluation might be indicated (see [Table 22.2](#)).

Rectal Medications

There are a variety of rectally administered medications that are used to trigger and sustain reflex defecation. These are usually inserted into the rectum to start bowel care expulsion and trigger gas and stool flow in as soon as a minute or as long as 30 minutes after liquification. Digital stimulation is then initiated to sustain the process, detect stool for manual disimpaction, or confirm the rectum is clear. The rationale for this prescription is to use the least irritating and most easily inserted and retained medication. Typically used suppositories include glycerin vegetable oil–based bisacodyl (Dulcolax) and polyethylene glycol–based (PGB) bisacodyl (Magic Bullet). Options for mini-enema–triggered bowel care include small-volume phosphor-soda enema, bisacodyl enema, and docusate mini-enema (Enemeez, formerly Theravac). Clinical efficacy studies have been conducted in attempts to measure efficiency and effectiveness of bowel care using various rectally administered triggering medications. In a randomized blinded study, hydrogenated vegetable oil–based (HVB) bisacodyl suppositories, PGB suppositories, and the docusate (Theravac) mini-enema were compared. Subjects with supraconal NBD were studied using the Events and Intervals of Bowel Care.⁶⁰ Bowel care trials with PGB bisacodyl suppositories showed an average time to defecation of approximately 22 minutes and a total bowel care time of 50 minutes. This was much shorter than the average time to defecation of 40 minutes and total bowel care time of 85 minutes observed with HVB bisacodyl suppository trials.

The use of mini-enemas had similar efficiency as PGB suppositories. Theravac has since been replaced by Enemeez. Results vary in individual patients and the use of digital stimulation alone is sufficient for many persons with SCI and supraconal NBD.

In supraconal NBD, digital stimulation to induce defecatory reflexes should be favored over manual disimpaction because the latter can easily result in inadvertent overstretching of the insensate and more delicate anal mechanisms of the neurogenic bowel. Local rectal stimulant suppositories and mini-enemas with bisacodyl or glycerin do not carry the same risk as oral stimulant medications and do not appear to lead to chronic inflammatory changes of the rectal mucosa.

Oral Medications for Bowel Care

When digital rectal stimulation, rectal medications, and rectal evacuation of stool are not sufficient to achieve the goals of the bowel program, oral medications are indicated (Table 22.3). Changes in medication with trials at active doses require at least 2 weeks or 4 bowel care sessions, at a minimum, to determine if a clinical response has started. Colon stimulants such as senna (Senokot) and bisacodyl (Dulcolax) activate colonic movement and increase fluid and electrolyte secretion in the GI tract. Osmotic laxatives promote water retention in the colon lumen. Osmotic laxatives that are commonly used can be classified into the following: (1) PGB solutions (e.g., Miralax), (2) nonabsorbable carbohydrates (e.g., lactulose [Cephulac]), (3) magnesium-containing solutions (milk of Magnesia, magnesium citrate), and (4) sodium phosphate-based products and stool softeners such as docusate (Colace). Determining the most appropriate oral medication or combinations thereof is usually done by trial and error based on type, dosage, quantity, and duration of efficacy. Constantly being aware of the goals of the bowel program assists with titration, revision, and timing of intake of oral medications. Approaches that appear effective initially need longer-term studies to verify their continued benefits due to a high incidence of late GI problems reported in an initially successfully managed SCI population.⁶⁷

Newer Oral Medications to Treat Constipation

Secretagogues. Secretagogues promote colonic transit and intestinal secretion by either affecting chloride channels or guanylate cyclase receptors.

Lubiprostone. Lubiprostone (Amitiza) increases intestinal fluid secretion by activating type 2 chloride channels, which promotes GI motility. This enhances intestinal and colonic transit and helps with passage of stool. It affects prostaglandin E receptors, which facilitate gastric and colonic muscle contraction and motility.^{37,98}

Linaclotide. Linaclotide (Linzess) is a 14-amino peptide agonist of guanylate cyclase-C (GC-C) receptor located on the luminal surface of intestinal epithelial cells. It increases cyclic guanosine monophosphate (cGMP), which triggers a signal transduction cascade activating the cystic fibrosis transmembrane conductance regulator. This results in secretion of fluid into the lumen and promotes intestinal transit. It is further shown to relieve abdominal pain by activating GC-C expressed on mucosal epithelial cells, resulting in the production and release of cGMP, inhibiting nociceptors.^{37,98}

Serotonin Agonists. Serotonin agonists affect 5-HT₄ receptors to facilitate colonic motility and intestinal secretion.

Prucalopride. Prucalopride is a selective 5-HT₄ receptor agonist that stimulates colonic transit and improves constipation by causing high-amplitude propagated contractions, hence enhancing segmental contractions.¹³⁴

Tegaserod. Tegaserod, a partial 5-HT₄ agonist, was removed from the US market when it was found to have increased risk for cardiovascular events.

μ-Receptor Antagonists. μ-Receptor antagonists selectively block μ-receptors peripherally in the GI tract, outside of the CNS.

For patients who have significant problems with constipation related to long-term opiate use due to its effects on μ-receptors, there are four medications currently available. Methylnaltrexone (Relistor, subcutaneous), alvimopan (Entereg), and naloxegol (Movantik) improve constipation without reversing analgesia or prompting opioid withdrawal.^{21,126} More recently, naldemedine (Symproic in the US, which is the only agent to bind to mu, kappa, and delta, has been shown to be effective compared to placebo.¹³⁶

Physical Interventions

Bowel Irrigation

Transanal irrigation with a unique enema system that includes a pump and a rectal balloon catheter (Peristeen) was found to be helpful in one large multisite randomized controlled study by improving constipation, reducing incontinence frequency, reducing total time for bowel care, reducing GI symptoms, and improving quality of life in individuals with SCI as compared with the regular bowel program.^{24,25,36,76,112} Subsequent studies further demonstrated lower costs of care,^{23,25} reduced or discontinued use of medications,^{33,76} and prolonged successful outcomes. Continued use of the device can successfully prevent constipation and/or FI and other symptoms.^{36,38,76}

Pulse irrigation enema (PIE) of the bowel has been shown to be a safe and effective method for relieving fecal loading and impactions by delivering intermittent, rapid pulses of warm water into the rectum for bowel cleansing and to stimulate rectal motility. However, the device is large and bulky and requires huge disposable bag collection systems.^{76,112}

Biofeedback and Pelvic Floor Physical Therapy

When neurogenic bowel deficits are incomplete and some degree of control and sensation is present, pelvic floor physical therapy and use of biofeedback might offer a means of enhancing the patient's residual sensory and motor abilities. This typically requires several sessions of therapy.¹⁷ Patients with SCI and NBD showed improvement in their sensory and motor anorectal function, including lowering of first sensation threshold and more effective balloon expulsion. They had improved symptom scores and positive effect of NBD on quality of life.⁹⁶ For individuals with a good degree of volitional EAS activation and some degree of anorectal sensation, biofeedback, and pelvic floor physical therapy can be a tool to help restore not only social continence but even normal defecatory control.^{96,140}

Abdominal Massage

Abdominal massage has been shown to promote better bowel movements, relieve abdominal distention, and improve colonic transit time in people with SCI. This was demonstrated in 24 subjects who received 15 minutes of abdominal massage from the cecum, throughout the colon, and to the rectum for 15 days.^{8,76} Constipation score was significantly improved in multiple sclerosis patients who received abdominal massage for 4 weeks, although there were no changes in the NBD or quality of life scores. Long-term improvement was not demonstrated when treatment was stopped after 4 weeks.^{25,97}

TABLE 22.3 Luminally Acting Agents for Constipation

Agent Category	Mechanism of Action	Clinical Considerations
Currently Available Agents		
Bulk laxatives (e.g., soluble fiber [psyllium, methylcellulose, calcium polycarbophil, partially hydrolyzed guar gum, wheat dextrin] and insoluble fiber [bran, flaxseed, rye])	Increases stool water content to soften stool; increased stool mass might stimulate peristalsis	Use in mild constipation; soluble fiber is more effective than insoluble fiber; psyllium and ispaghula husk most studied; avoid when dyssynergia present
Surfactant laxatives (e.g., docusate sodium, docusate calcium)	Anionic detergents lower the surface tension of stool; allows water to penetrate stool	Use in mild constipation; psyllium is more effective than docusate
Osmotic laxatives (e.g., PEG, lactulose, sorbitol, magnesium salts)	Generation of an osmotic gradient in gut lumen; promotes movement of water into lumen; luminal water softens stool and stimulates secondary peristalsis	PEG and lactulose effective for intermittent and chronic constipation; PEG is more effective than lactulose; might not benefit pain in IBS-C; avoid use of magnesium in patients with renal dysfunction
Stimulant laxatives (e.g., diphenylmethanes [bisacodyl, sodium picosulfate], anthraquinones [senna, cascara], misoprostol, castor oil)	Direct colonic wall irritant; stimulation of sensory nerves on colonic mucosa; possible inhibition of water absorption; prostaglandin-induced effects on motility and secretion with misoprostol	Efficacy for intermittent constipation; diphenylmethanes effective for chronic constipation; long-term safety not established
Chloride channel activation (e.g., lubiprostone)	Secretion of chloride ions into intestinal lumen through direct activation of Cl-C2 chloride channels on enterocytes; results in passive movement of sodium and water into intestine	Short- and long-term efficacy and safety data in chronic constipation and women with IBS-C; main adverse event is dose-dependent nausea
Probiotics (e.g., <i>Bifidobacterium lactis</i> , <i>Lactobacillus paracasei</i>)	Hypothesized effects on gut transit and secretion through alteration of gut microbiota	Possible role in chronic constipation and IBS-C; no long-term efficacy or safety data; quality control issues (regulated as food additives not drugs)
Emerging Agents		
Chloride-channel activation (e.g., linaclotide, plecanatide)	Activation of guanylate cyclase-C receptor generating cGMP; secretion of chloride ions into intestinal lumen through cGMP mediated activation of CFTR; results in passive movement of sodium and water into intestine; inhibition of visceral pain fiber firing by cGMP in animals	Effective in chronic constipation and IBS-C in phase II (linaclotide and plecanatide) and phase III clinical trials (linaclotide); main adverse event is diarrhea
Bile-acid analogs (e.g., chenodeoxycholic acid)	Increases colonic motor activity; increases luminal secretory activity	Effective in IBS-C as shown in phase II trial; risk of abdominal pain and/or cramps
Inhibitors of bile-acid resorption (e.g., elobixibat)	Partial inhibition of ileal bile acid transporter; increases colonic bile acid concentrations promoting colonic motility and secretion	Effective for chronic constipation shown in phase II trial; dose-dependent abdominal pain reported
Peripherally acting μ -opioid receptor antagonists (e.g., methylnaltrexone, alvimopan)	Improve constipation without reversing analgesia and or prompting opioid withdrawal by peripherally acting μ -opioid receptor antagonists, which selectively block μ -receptors outside of the CNS	Effective in controlling constipation in chronic opioid users; methylnaltrexone (6 RCTs), alvimopan (four RCTs) were shown to be superior to placebo: adverse effects abdominal pain and diarrhea.

CFTR, Cystic fibrosis transmembrane conductance regulator; cGMP, cyclic guanosine monophosphate; CNS, central nervous system; IBS-C, constipation-predominant irritable bowel syndrome; PEG, polyethylene glycol; RCT, randomized controlled trial.

Modified from Menees S, Saad R, Chey WD. Agents that act luminally to treat diarrhoea and constipation. *Nat Rev Gastroenterol Hepatol*. 2012;9(11):661–674.

Functional Electrical Stimulation/ Functional Magnetic Stimulation

Use of an abdominal belt with electrodes for electrical stimulation of the abdominal muscles in people with tetraplegia was shown to improve colonic transit time with overnight use in a good randomized control trial.^{25,73,76} Another study showed similar results in

reducing colonic transit time with electrical stimulation of the abdominal muscles for 25 minutes, 5 days per week for 8 weeks compared with placebo.^{55,76} Comparatively, individuals with SCI were shown to have shorter colonic transit times with functional magnetic stimulation of the abdominal muscles, but better-quality studies are necessary.^{76,86,87}

Surgical Options

Gastrostomies and Enterostomies

Optimization of nutrition via gastrostomies and enterostomies has been found to be beneficial in patients with gastroparesis, intestinal, or colonic pseudo-obstruction. It not only helps to prevent malnutrition, but it also alleviates discomfort from nausea, vomiting, and abdominal distention. For people on parenteral nutrition, a venting enterostomy provides symptomatic relief from gaseous distention and bloating.

Surgeries for Chronic Intestinal or Colonic Pseudo-Obstruction

During acute subocclusive episodes, colonoscopic decompression is successful in 75% to 90% of cases where nasogastric or rectal tubes were ineffective. A colonic decompression tube with suctioning can be used for those requiring repeated decompression. Decompression of the GI tract also improves forward motility and coordinated propulsion. For cases in which conservative treatment has failed, subtotal colectomy with ileorectostomy has been found to be the most effective for chronic colonic pseudo-obstruction.

Pelvic Floor Sling

Sacral nerve deficits due to trauma, tumor-related mass effect, or other cause can interfere with the action of the puborectalis, levator ani, and EAS (see Fig. 22.3). The resulting pelvic floor descent impairs the protective puborectalis sling angle and decreases the efficacy of protective EAS contractions. Some patients have benefited from transposition of innervated gracilis, adductor longus, gluteus maximus, or free muscle graft palmaris longus to replace puborectalis function and restore the acute anorectal junction angle. Chronic electrical stimulation to enhance development of fatigue-resistance type I fibers is used with these transplants. Sensory deficits are not improved, but continence is somewhat restored with the ability to inhibit defecation if some degree of sensation remains.

Electroprosthesis

Stimulation of anterior sacral roots (S2, S3, and S4) for voluntary micturition has been accomplished via stimulator. Surgically placed selective stimulation of S2 tends to promote nonperistaltic, low-level electrical impulses to the sacral plexus, influencing the anal canal, colon, and pelvic floor musculature. Stimulation of S3 causes occasional high-pressure peristaltic waves, especially with repetitive stimulation and stimulation of S4 increases both rectal and anal tone. However, consistent changes in anal resting pressure and other manometric outcome measures have not been reported in the literature.

The sacral anterior root stimulator (SARS), which is controlled by an external remote, has been used in individuals with SCI to manage neurogenic bladder with electrodes surgically placed in the S2, S3, and S4 anterior nerve roots. The stimulator delivers short, high-voltage electrical stimulation to the bladder for emptying. In addition, the sacral anterior root stimulator has produced good results for neurogenic bowel management^{30,63} by facilitating colonic motility and triggering spontaneous bowel evacuation,^{22,25,76,138} reducing constipation and duration of bowel care,^{22,25,63,76,138} preventing autonomic dysreflexia during bowel care, enhancing quality of life,^{25,63,76} and reducing need for caregiver assistance.^{76,92} FI and quality of life were also shown to improve with sacral nerve root stimulation in patients with cauda equina syndrome.^{48,76}

Antegrade Continence Enema

The options for trial of the antegrade continence enema (ACE) should be considered in clinical scenarios of prolonged bowel care time, recurrent fecal impactions, or poor or intermittent response to rectal medications to initiate bowel care. This is an alternative method of antegrade enema delivery that requires the surgical construction of a catheterizable appendicocostomy stoma. The appendix and right colon are mobilized through a horizontal right lower quadrant incision and brought against the abdominal wall. The tip of the appendix is then amputated, and the opening into the appendix lumen is modified into a catheterizable stoma on the abdominal wall. This procedure can now also be performed laparoscopically.¹²⁴ This stoma can be catheterized and infused with 200 to 600 mL of tap water to trigger a propulsive colonic peristalsis and defecation within 10 to 20 minutes. Bowel care can then be additionally facilitated with digital stimulation in the usual fashion. This procedure has been utilized most commonly in children with meningomyelocele.

Colostomy

Colostomies are indicated in four general scenarios: (1) when conservative bowel program medical measures and training have failed, (2) when repetitive bowel impactions occur, (3) when pressure ulcers or other skin lesions occur that cannot be effectively healed because of frequent soiling, or (4) when intrinsic bowel deficits exist such as in Hirschsprung disease, Chagas disease, and colonic atresia. Colostomies have been demonstrated to be beneficial in various ways in multiple studies and systematic reviews in people with SCIs.⁶ It has been shown that colostomies promote ease in bowel care by simplifying and decreasing time spent on bowel management,^{14,65,76,118} reduce bowel symptoms,^{76,118} mitigate episodes of incontinence, lower rate of hospital admissions from GI problems,^{76,118} encourage independence, and raise quality of life.^{65,76,118} Following colostomy, patients have wished they had the colostomy done sooner.^{14,76} A sigmoid colostomy (left sided, more distal) is typically recommended to allow absorption of fluid, preserve hydration, and have better formed stools.^{76,120} High stoma placement allows for better visualization.

Complications

For people with gastroparesis or intestinal pseudo-obstruction, the primary complication is chronic malnutrition, dehydration, and electrolyte imbalance. It is important to optimize nutrition and hydration and to monitor weight, vitamin, and electrolyte levels regularly. Oral and enteral feedings must provide adequate calories, protein, electrolytes, and vitamins. The liquid formula should be low in fat, fiber, and lactose; high in polypeptides and hydrolyzed protein; and received in small, frequent amounts.

In acute subocclusive episodes of colonic pseudo-obstruction, it is important to achieve decompression as soon as possible. Failure to do so can result in progressive distention and cecal ischemia, causing perforation. There is increased risk of perforation in cecal diameters over 12 cm. Colonic perforation requires emergent exploratory laparotomy and intestinal resection. Mortality is approximately 36% to 44% and is determined by age, diameter of the cecum, delay of decompression, and comorbidities.

Significant bowel complications requiring medical treatment or lifestyle alterations are reported by 27% of persons with SCI by

^aReferences 14,25,27,76,120.

5 years or greater beyond their injury, even if bowel management was satisfactory during the first 5 years. More than 80% of persons with SCI reported bowel impactions, and 20% had chronic bowel impaction and DWE problems.⁶⁰ Impactions have been reported to be complicated by perforation or even death. Impactions have a morbidity ranging up to 6% in the “normal” population and are higher in the cognitively impaired elderly.¹³² Other late GI complications reported by patients with SCI include gastroesophageal reflux, premature diverticulosis, and autonomic dysreflexia. Morbidity from colonic perforation by enema use has also been reported.¹²⁴

Hemorrhoids are more symptomatic when patients have intact sensation, but in one study rectal bleeding due to hemorrhoids was reported by 74% of patients with SCI.¹²⁴ Hemorrhoids develop as a result of frequent high pressures in the anorectal marginal veins and are associated with constipated hard stool passage. Stool softening is the best preventative and chronic treatment measure, but it should be balanced with the requirement to modulate stool consistency to maintain continence.

An overstretched patulous noncompetent sphincter associated with rectal prolapse often is the end result of chronic passage of very large hard stools through a weakened anorectal mechanism in infraconal NBD. Overdistention of the weakened neurogenic anal mechanism should be avoided by the use of stool softening and gentle care to break up stool and dilate the sphincter whenever manual disimpaction is required. Although the anus can be significantly dilated to accommodate two fingers for breaking up low impactions, anorectal overdistention has been hypothesized to lead to atonic segments similar to bladder overdistention.

Autonomic dysreflexia occurs in patients with SCI and lesions at or above the midthoracic region. Fecal impaction is a common and potentially dangerous cause of autonomic dysreflexia because of the substantial time that can be required for its clearance. If manual disimpaction is required, lubrication with lidocaine gel is recommended to decrease additional nociceptive sensory input from the richly innervated anal region.

Bloating and abdominal distention are common complaints of patients with NBD. These complaints can be reduced in patients with SCI by increasing the frequency of bowel care, decreasing gas forming foods and drinks, and taking medications that can help gaseousness such as simethicone. This complaint can be especially severe in those with hyperactive EAS protective responses to rectal distention, which can preclude the passing of flatus. Digital release of flatus might be required in addition to diet modification to eliminate foods that produce excessive gases. The workup should also include assessment for any contributing aerophagia (air swallowing).

Treatment Outcomes

Bowel habituation training in children with myelomeningocele by means of suppositories, digital stimulation, or both, resulted in 83% of compliant patients having less than one incontinent stool per month.⁷² The continence catheter enema, which has a distal rectal balloon to avoid immediate enema expulsion, when used daily or every other day, reduced FI to fewer than three episodes per month in children with myelomeningocele.⁹⁵

Although all patients with complete SCI have episodic FI, this is a chronic problem for only 2%. DWE appears to be a progressive problem that develops 5 years or more after SCI. This is rarely reported after training in the first 4 years but occurs in 20% by a

mean of 17 years after injury. GI problems in SCI are not merely nuisances and account for 10% of SCI late mortality.²⁹

Patients with multiple sclerosis, Parkinson disease, or muscular dystrophy have also been helped by methods to enhance bowel storage or elimination in the setting of deteriorating neuromuscular and anorectal function.⁹⁴ Colostomy can provide a means of achieving social continence in these patients. Colostomies require expert or self-management and have implications such as embarrassing gas problems, appliance loosening and leakage, and cosmetic difficulties.

Patients who develop social bowel continence are able to venture into public without fear of malodorous embarrassment and unpredictable social disasters that humiliate as well as require substantial clean-up time. When fear of such adverse events persists, full social and vocational reintegration is impeded. A major hurdle that many patients with neuromuscular compromise can overcome is control of the seemingly automatic neurogenic functions of defecation and bowel elimination. Such patients should not needlessly suffer because inadequate attention has been paid by care providers to this potentially functionally impairing and socially disabling deficit.

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